

Alligator Harbor Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 05 May, 2026

Contents

Indicators	2
Nutrients	2
Total Nitrogen - Discrete	2
Total Phosphorus - Discrete	4
Water Quality	6
Dissolved Oxygen - Continuous	6
Dissolved Oxygen - Discrete	8
Dissolved Oxygen Saturation - Continuous	10
Dissolved Oxygen Saturation - Discrete	12
Salinity - Discrete	14
Salinity - Continuous	16
Water Temperature - Continuous	18
Water Temperature - Discrete	20
pH - Continuous	22
pH - Discrete	24
Specific Conductivity - Continuous	26
Water Clarity	28
Turbidity - Continuous	28
Turbidity - Discrete	30
Total Suspended Solids - Discrete	32
Chlorophyll a, Uncorrected for Pheophytin - Discrete	34
Chlorophyll a, Corrected for Pheophytin - Discrete	36
Secchi Depth - Discrete	38
Colored Dissolved Organic Matter - Discrete	40

Indicators

Nutrients

Total Nitrogen - Discrete

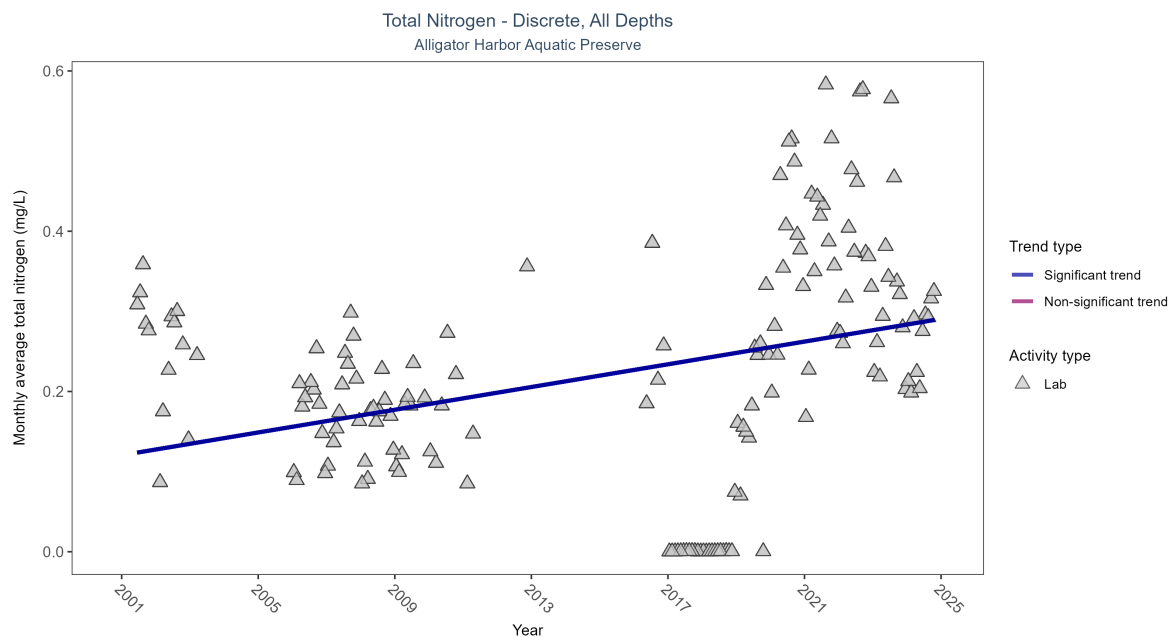


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	1625	19	2001 - 2024	0.2	0.27903	0.12051	0.00709	0

Monthly average total nitrogen increased by 0.01 mg/L per year.

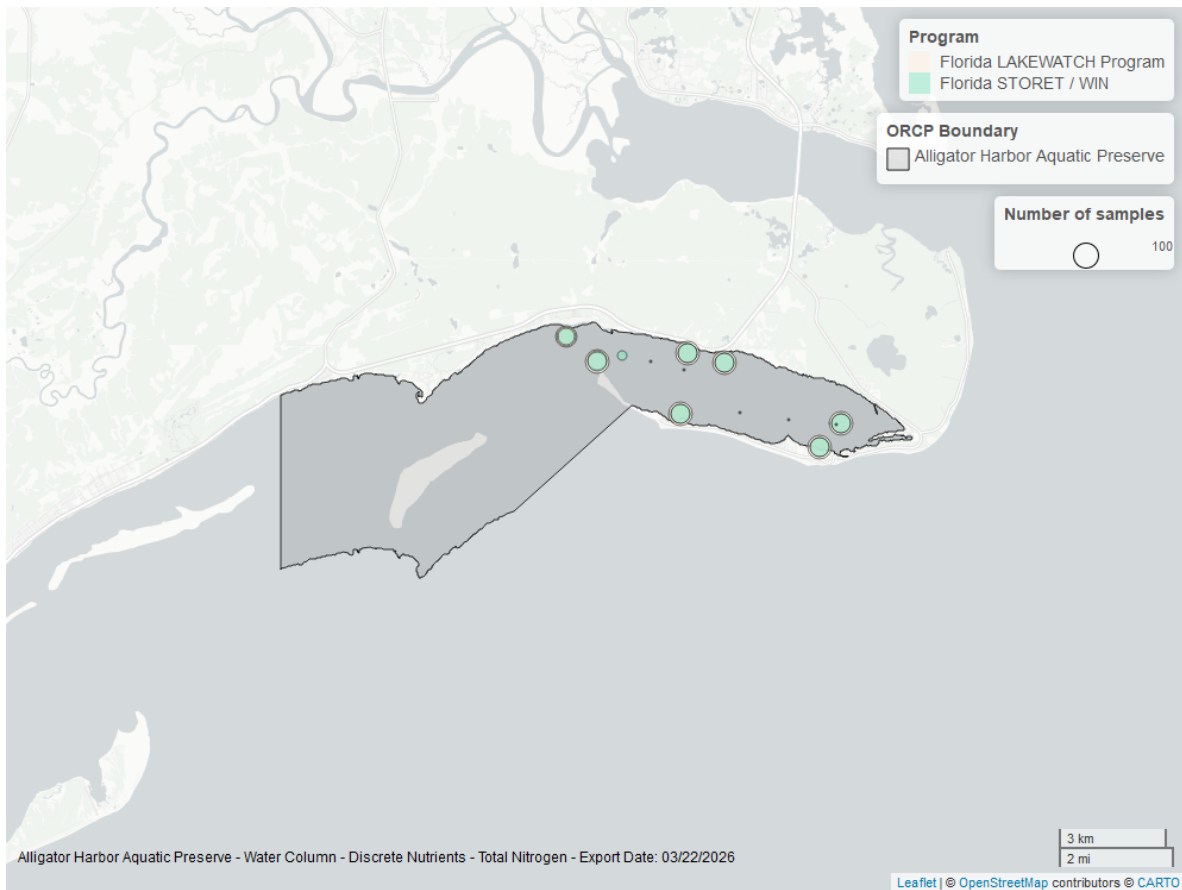


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

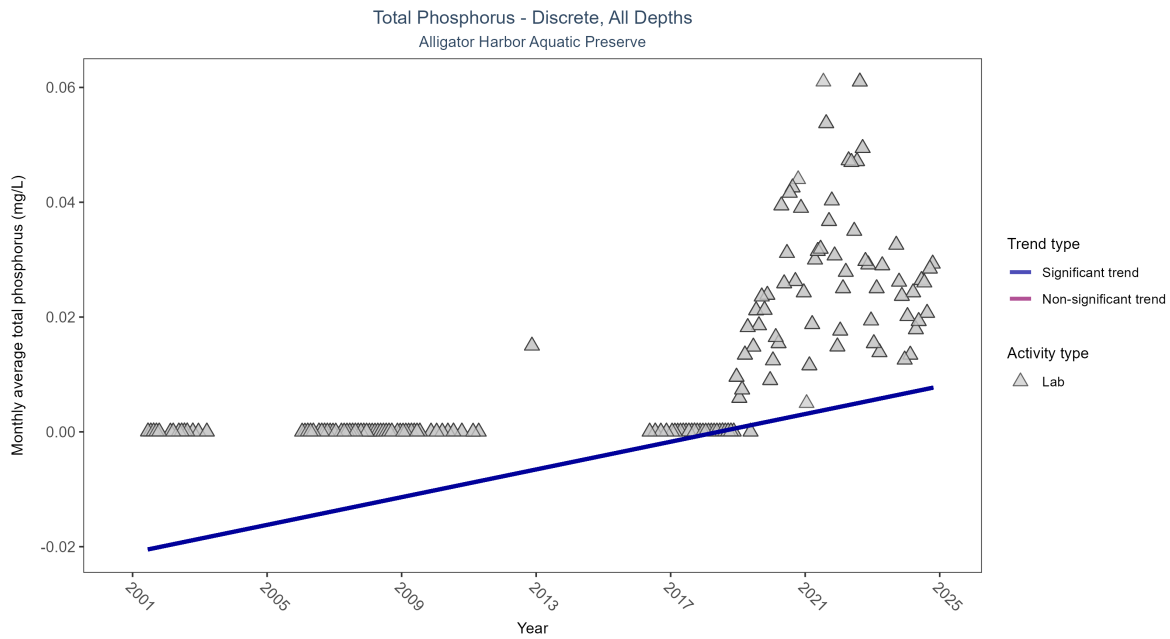


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	1130	19	2001 - 2024	0.00005	0.55833	-0.02103	0.00121	0

Monthly average total phosphorus increased by less than 0.01 mg/L per year.

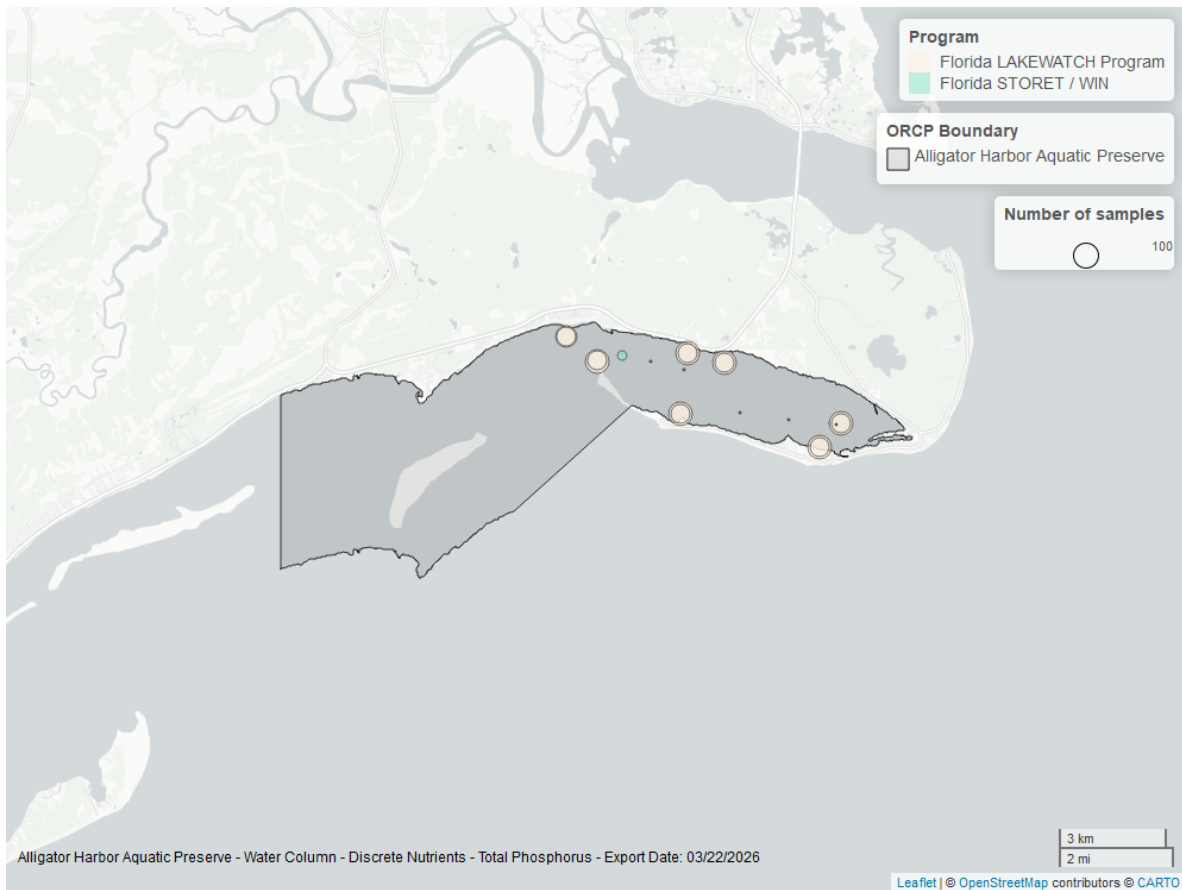


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

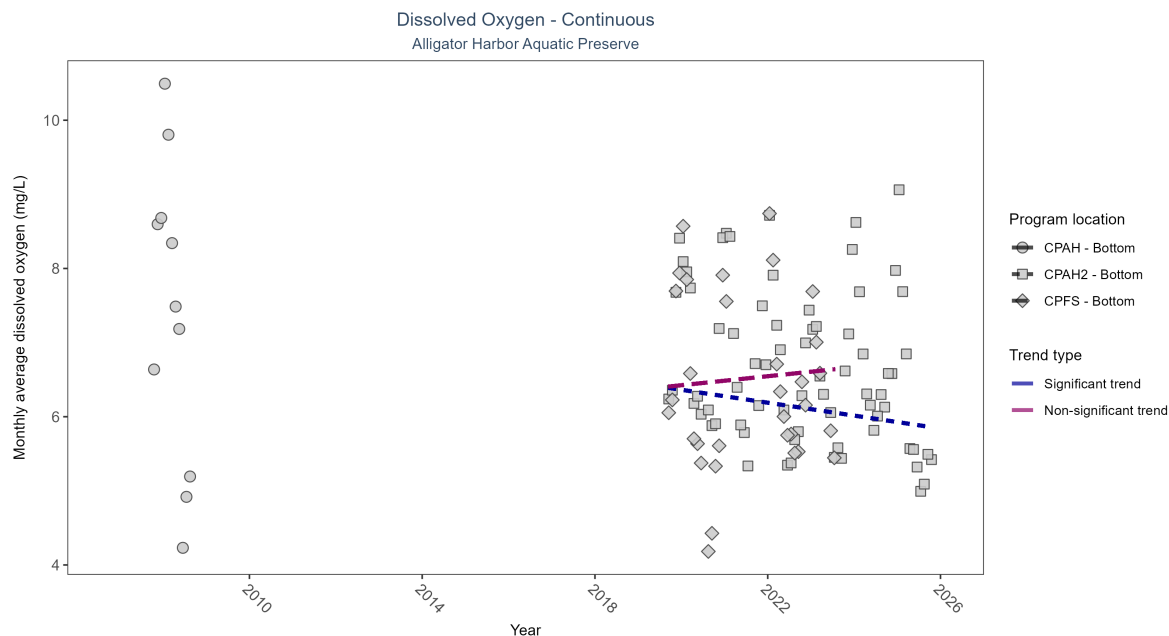


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data to calculate trend	11557	2	2007 - 2008	8.0	-	-	-	-
CPAH2	Significantly decreasing trend	175461	7	2019 - 2025	6.7	-0.24	6.45	-0.09	0.0221
CPFS	No significant trend	56853	5	2019 - 2023	6.4	0.12	6.36	0.06	0.7377

At one program location, monthly average dissolved oxygen decreased by 0.09 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at one location. There was insufficient data to fit a model for one location.

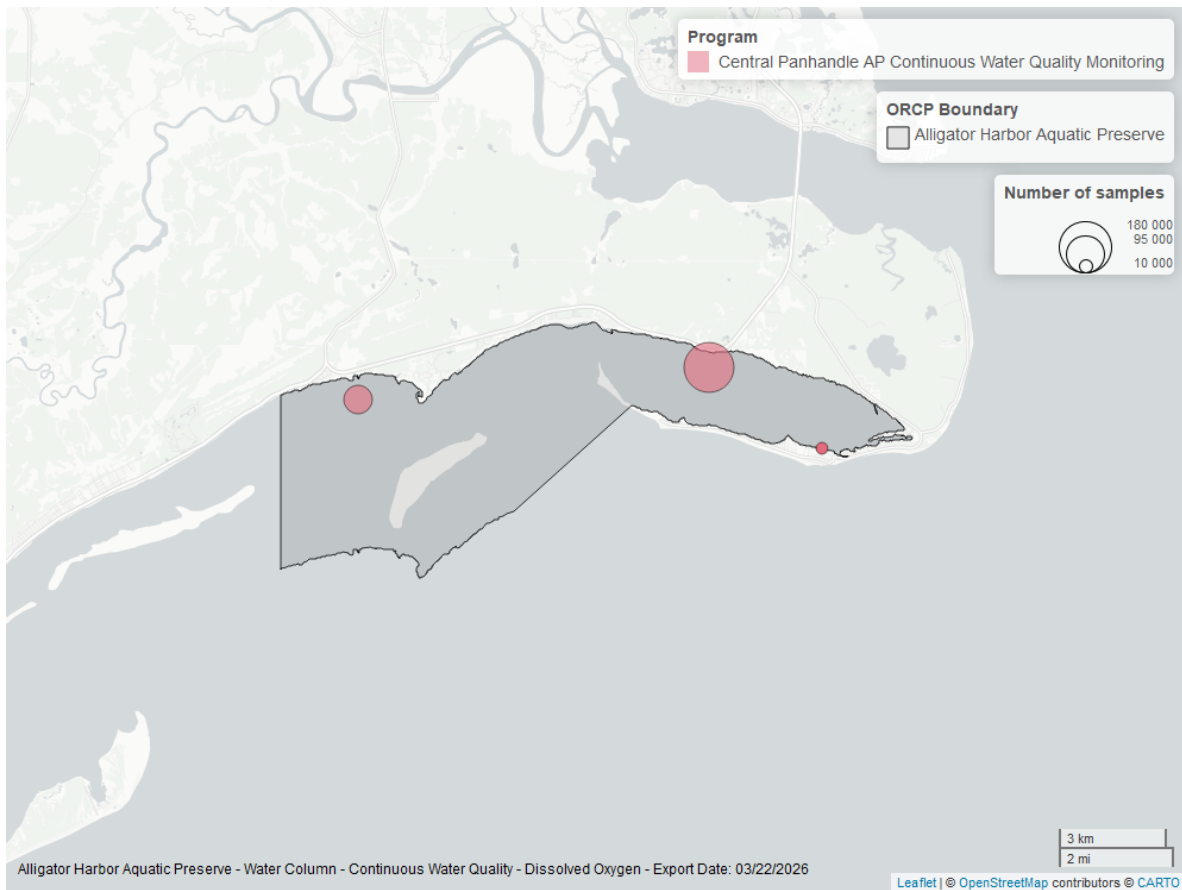


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

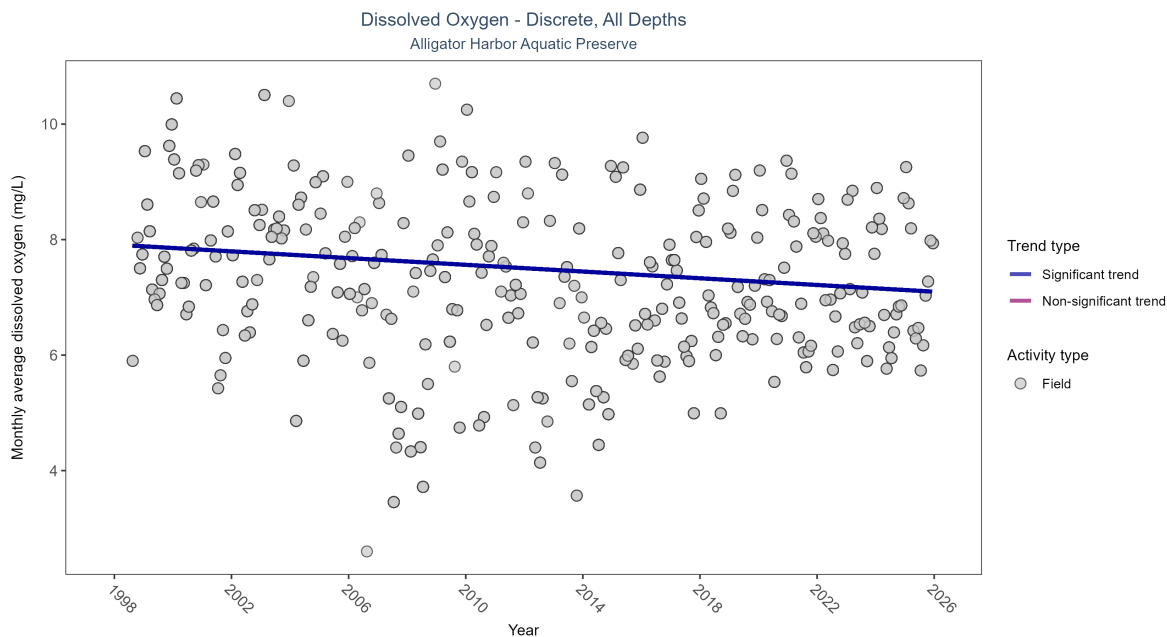


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	8423	28	1998 - 2025	7.2	-0.1683	7.91292	-0.02911	0

Monthly average dissolved oxygen decreased by 0.03 mg/L per year.

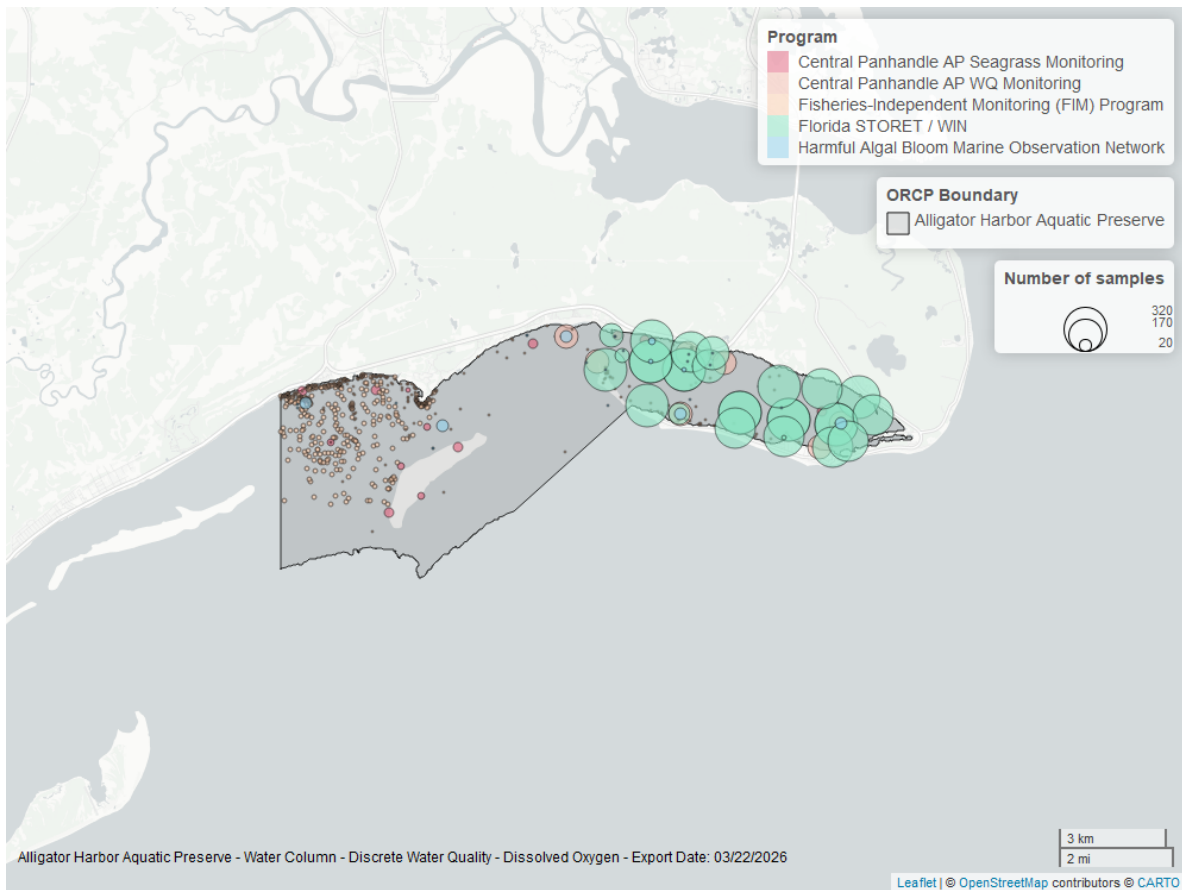


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

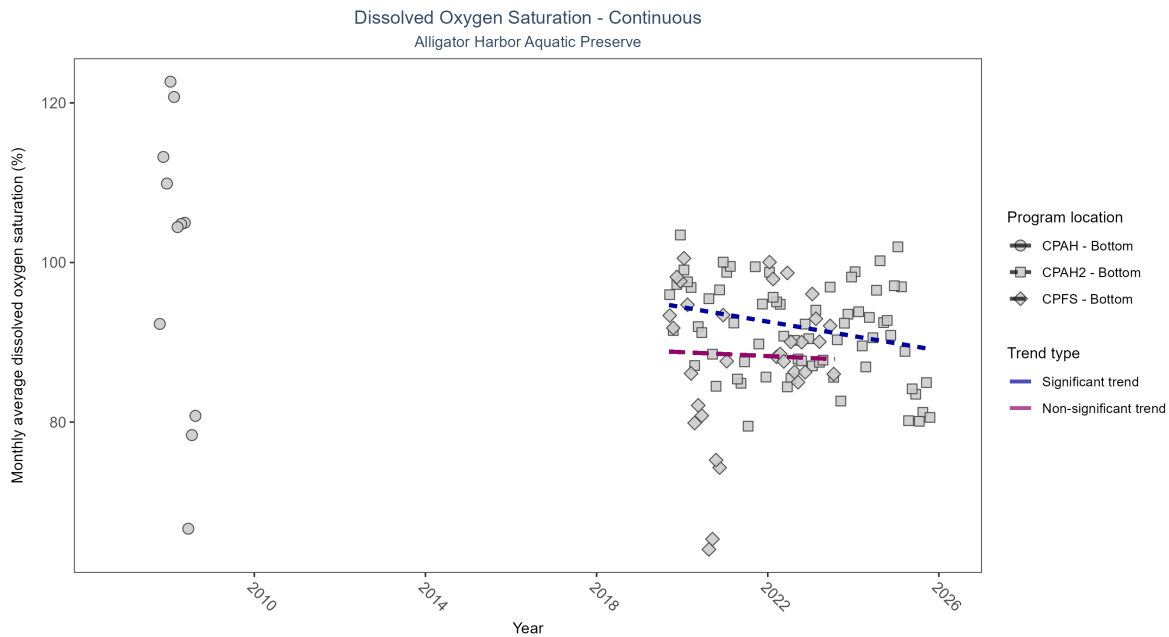


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data to calculate trend	11557	2	2007 - 2008	105.6	-	-	-	-
CPAH2	Significantly decreasing trend	175461	7	2019 - 2025	93.0	-0.27	95.3	-0.91	0.0089
CPFS	No significant trend	56928	5	2019 - 2023	90.7	0.02	89.01	-0.24	1

At one program location, monthly average dissolved oxygen saturation decreased by 0.90% per year. No detectable change in monthly average dissolved oxygen saturation was observed at one location. There was insufficient data to fit a model for one location.

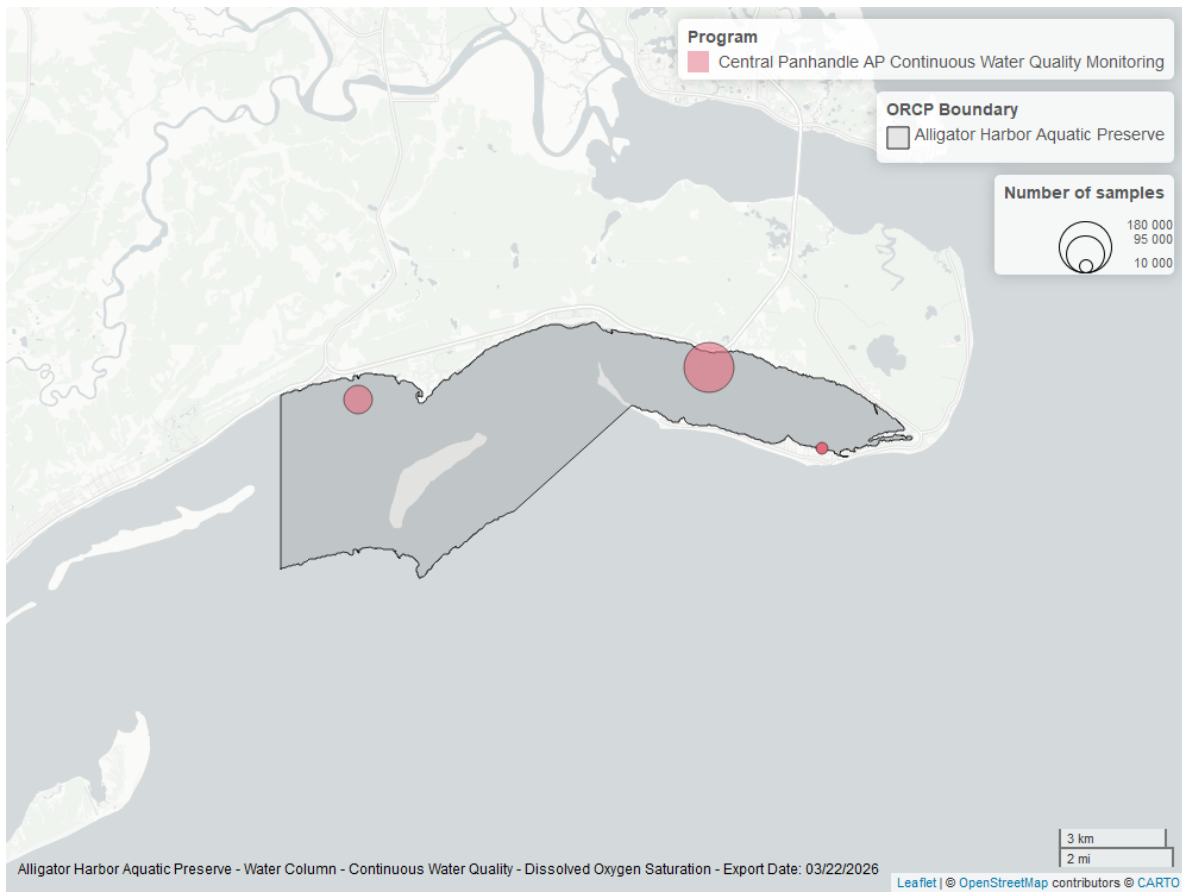


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

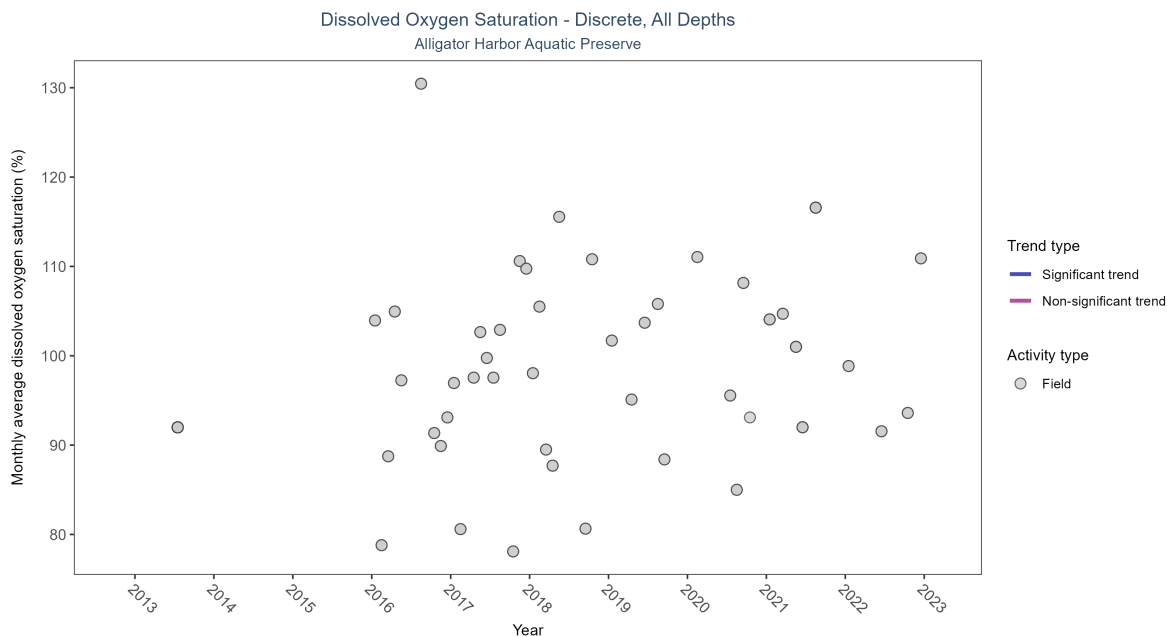


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Insufficient data to calculate trend	131	8	2013 - 2022	96.8	-	-	-	-

There was insufficient data to fit a model for dissolved oxygen saturation.

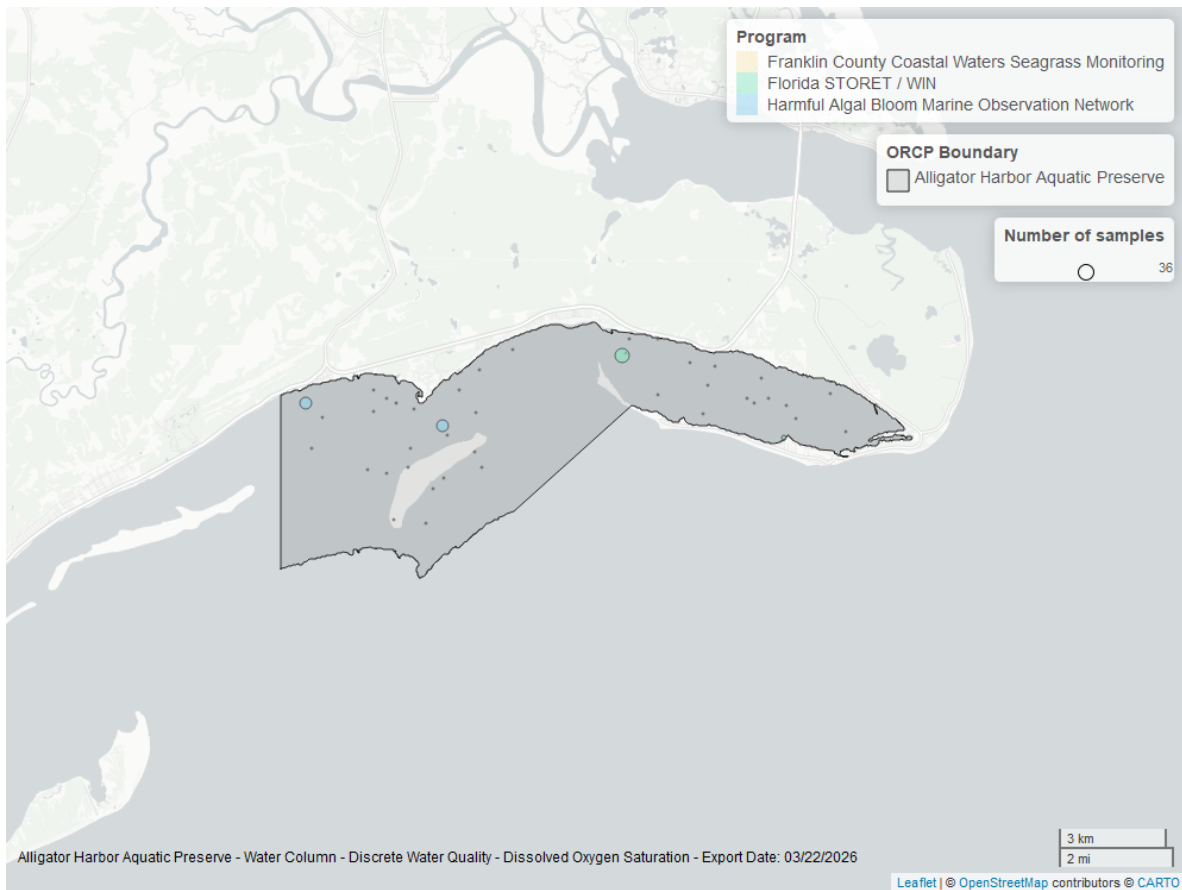


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

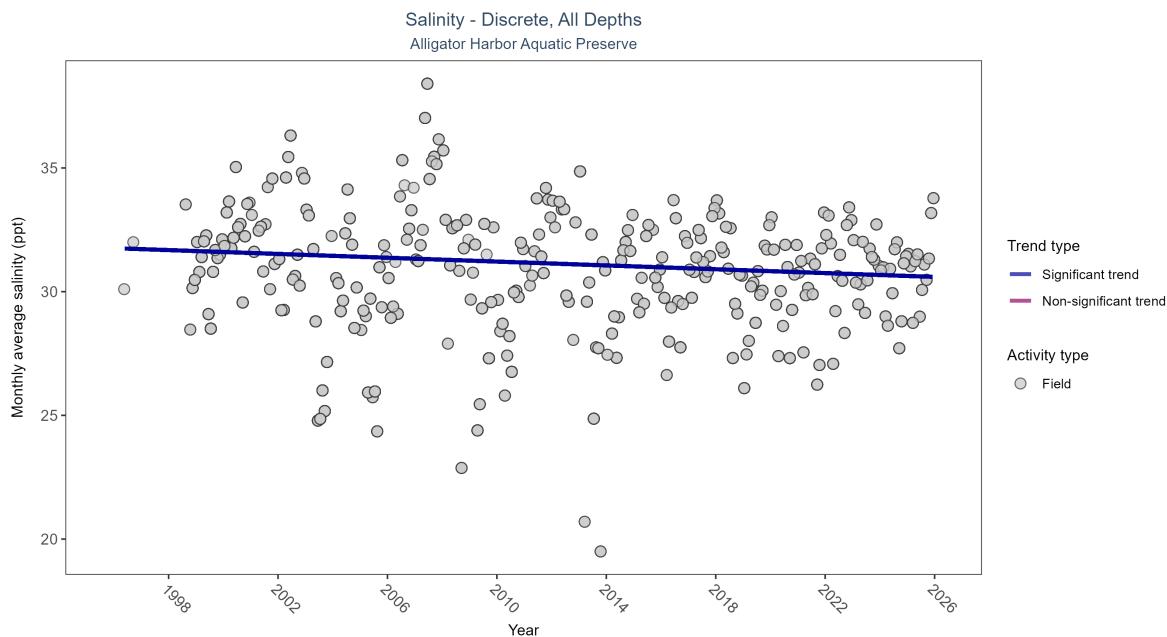


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Significantly decreasing trend	9675	29	1996 - 2025	31.11	-0.107	31.75772	-0.03871	0.0072

Monthly average salinity decreased by 0.04 ppt per year.

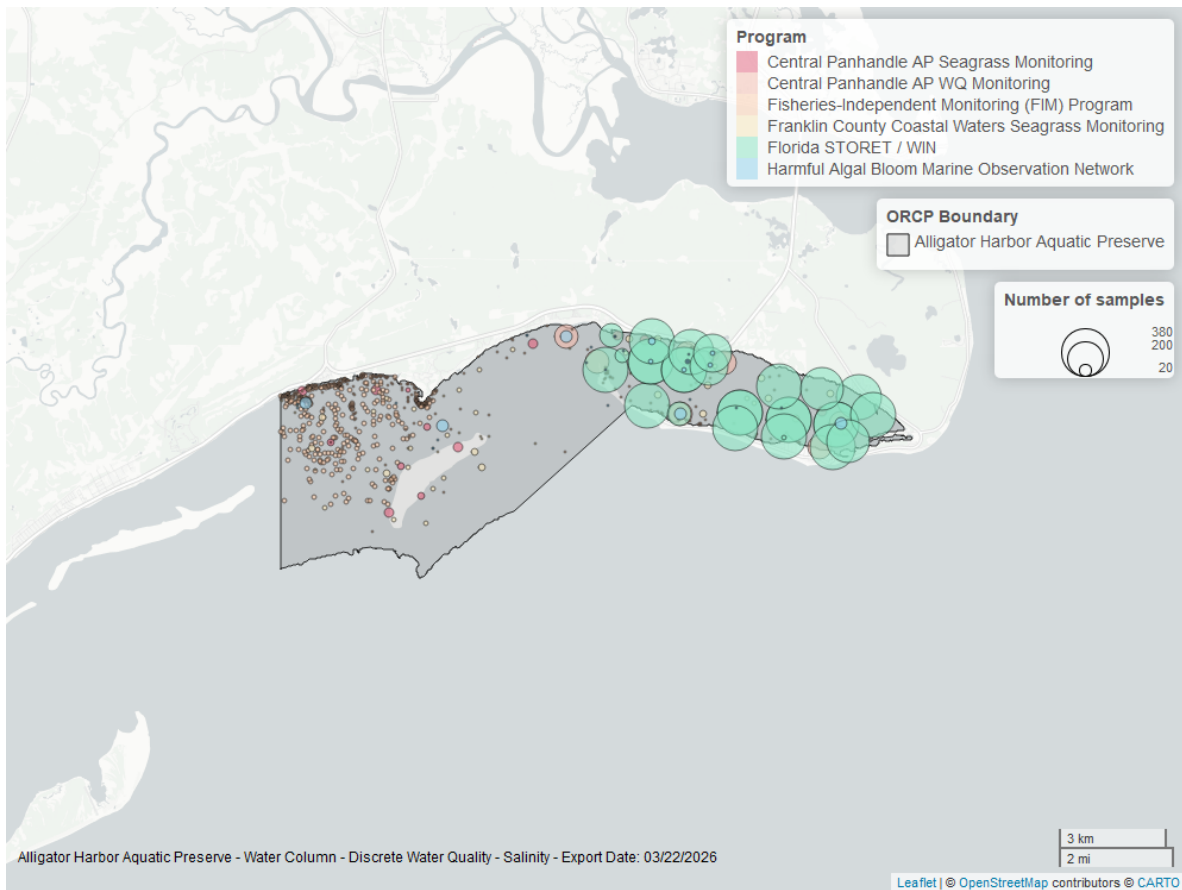


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

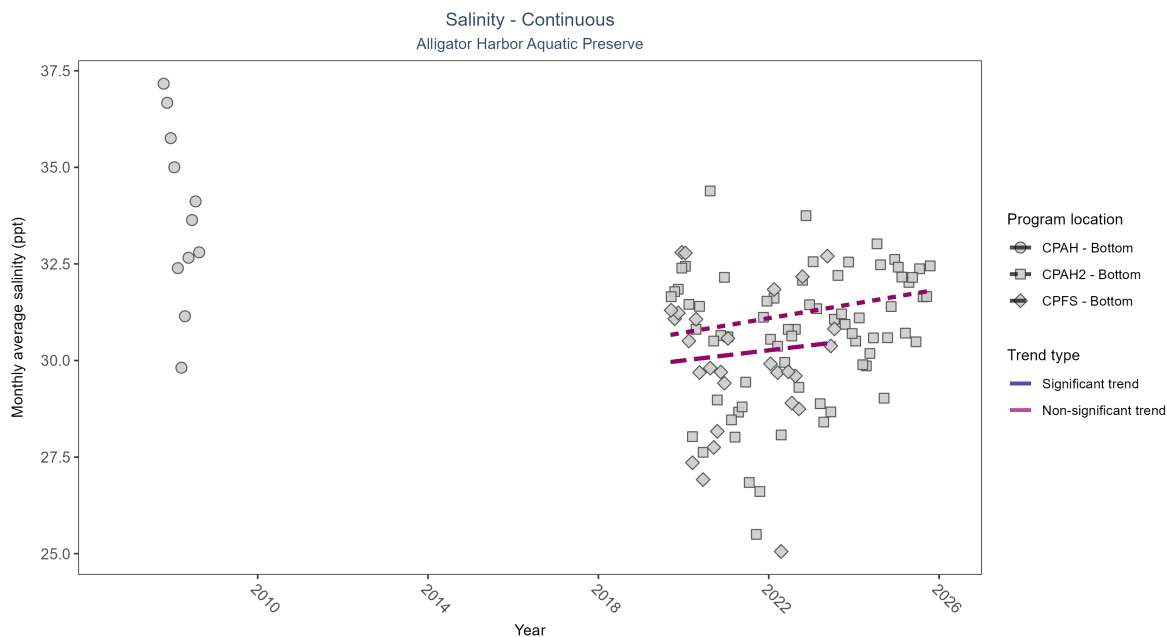


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data to calculate trend	13034	2	2007 - 2008	34.0	-	-	-	-
CPAH2	No significant trend	171623	7	2019 - 2025	31.0	0.18	30.53	0.19	0.1022
CPFS	No significant trend	60346	5	2019 - 2023	30.5	0	29.87	0.13	1

No detectable change in monthly average salinity was observed at two locations. There was insufficient data to fit a model for one location.

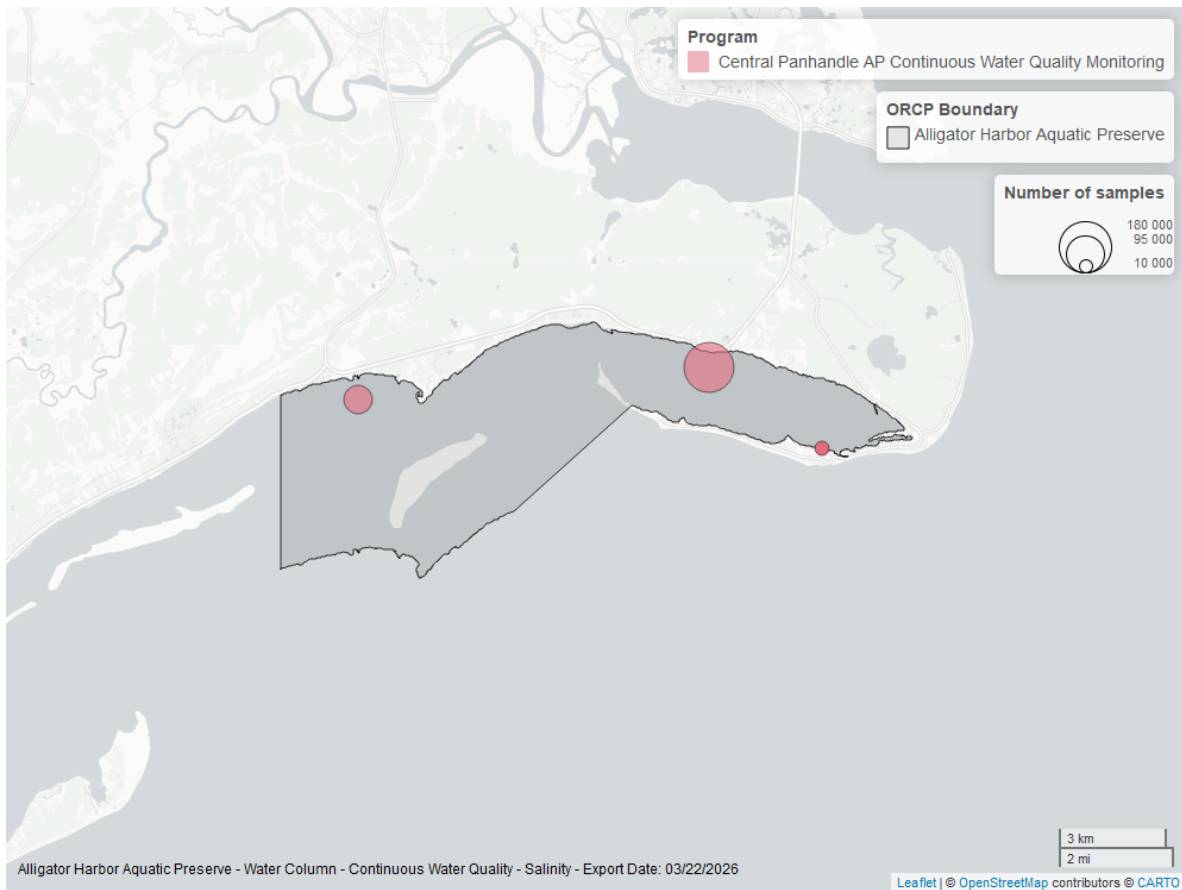


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

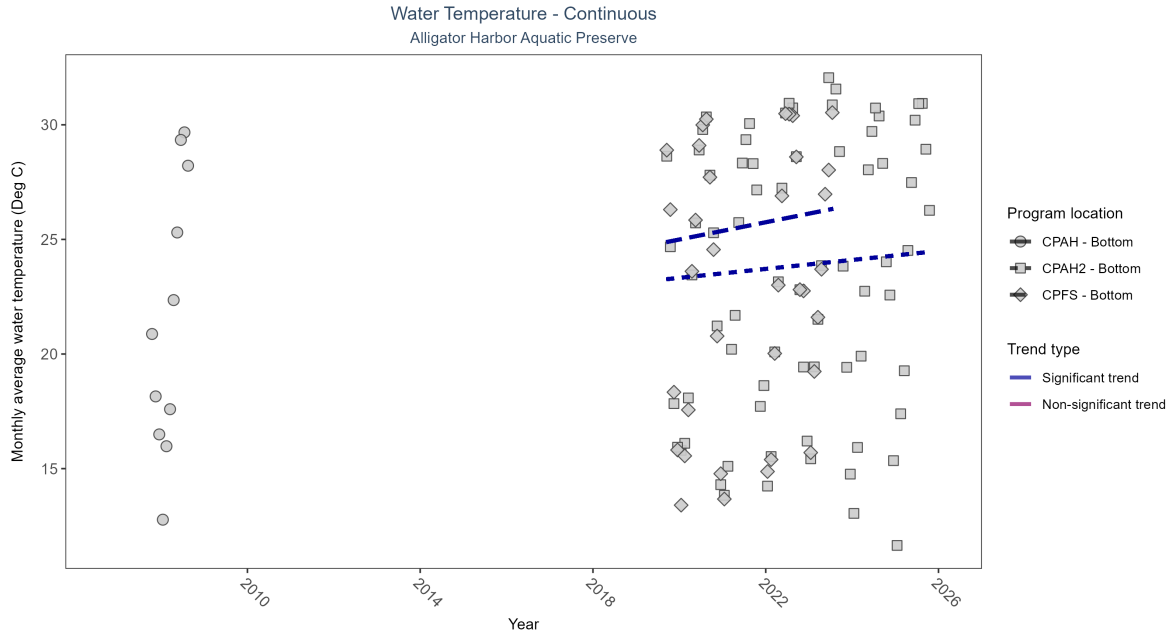


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data to calculate trend	13034	2	2007 - 2008	20.9	-	-	-	-
CPAH2	Significantly increasing trend	189604	7	2019 - 2025	23.7	0.22	23.13	0.2	0.0453
CPFS	Significantly increasing trend	80801	5	2019 - 2023	24.1	0.37	24.62	0.38	0.0341

At two program locations, monthly average water temperature increased by 0.2°C per year at one site and by 0.38°C per year at the other. There was insufficient data to fit a model for one location.

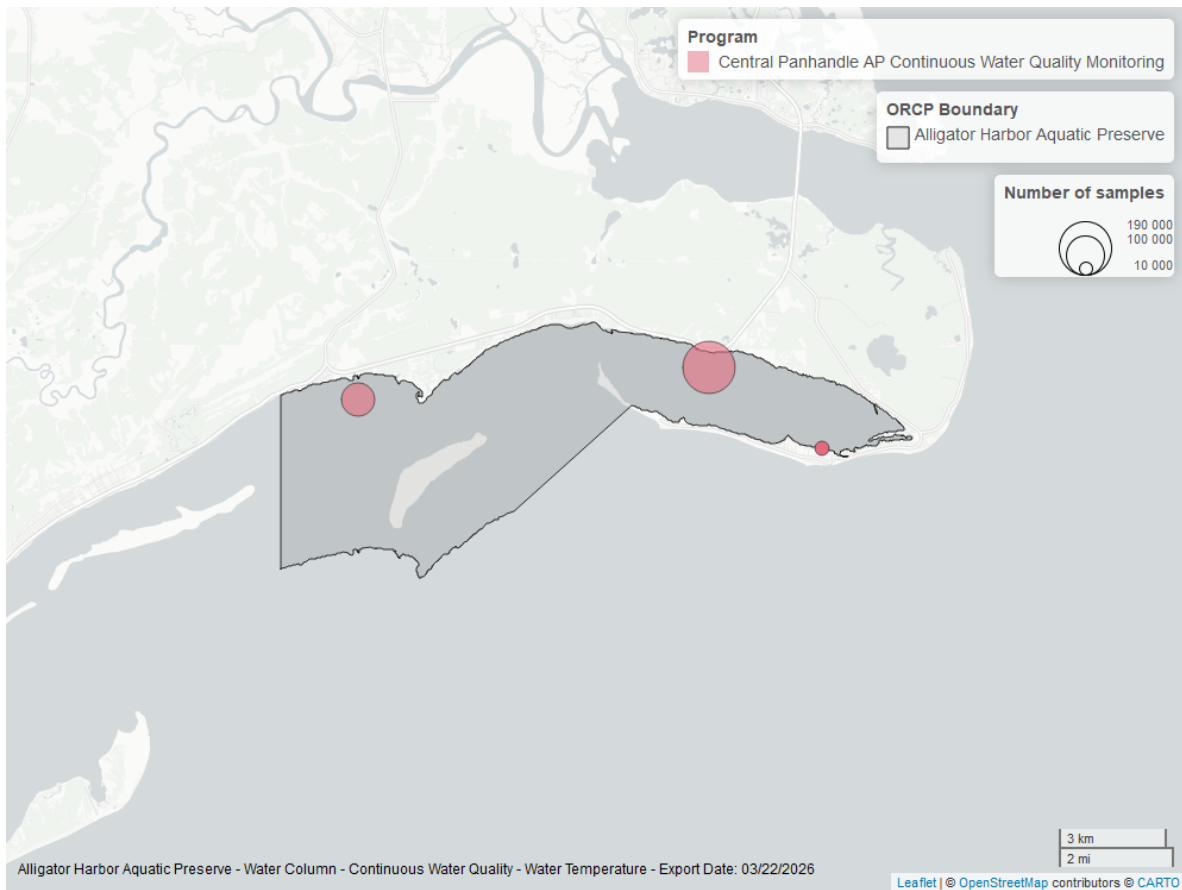


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

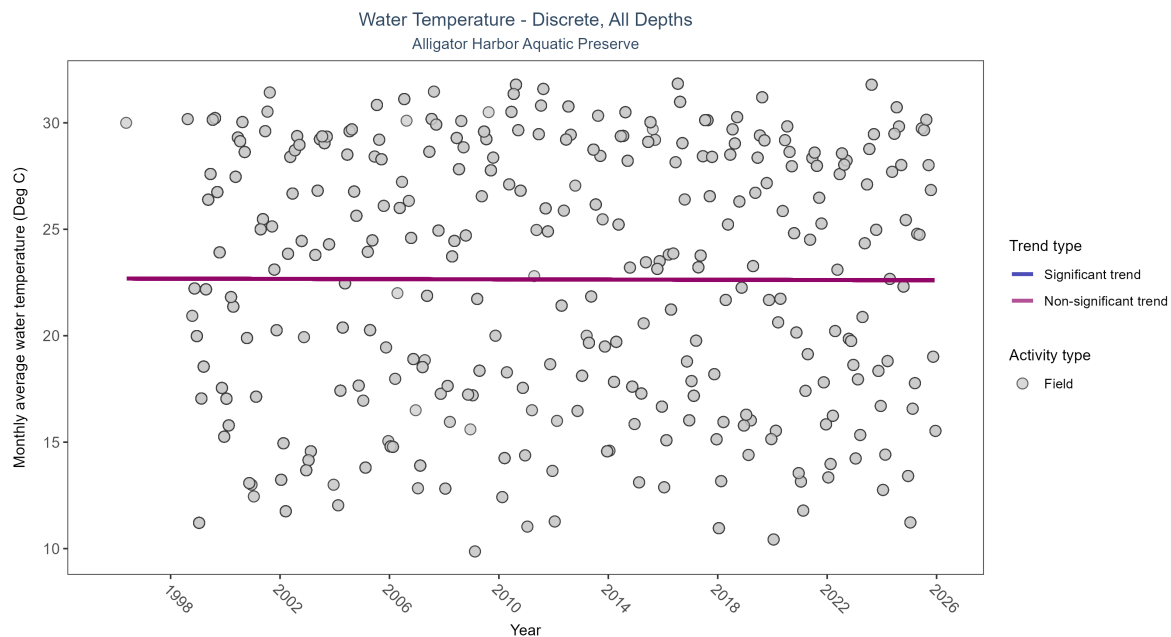


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	9916	29	1996 - 2025	25.9	-0.00609	22.68476	-0.00264	0.8755

Water temperature showed no detectable trend between 1996 and 2025.

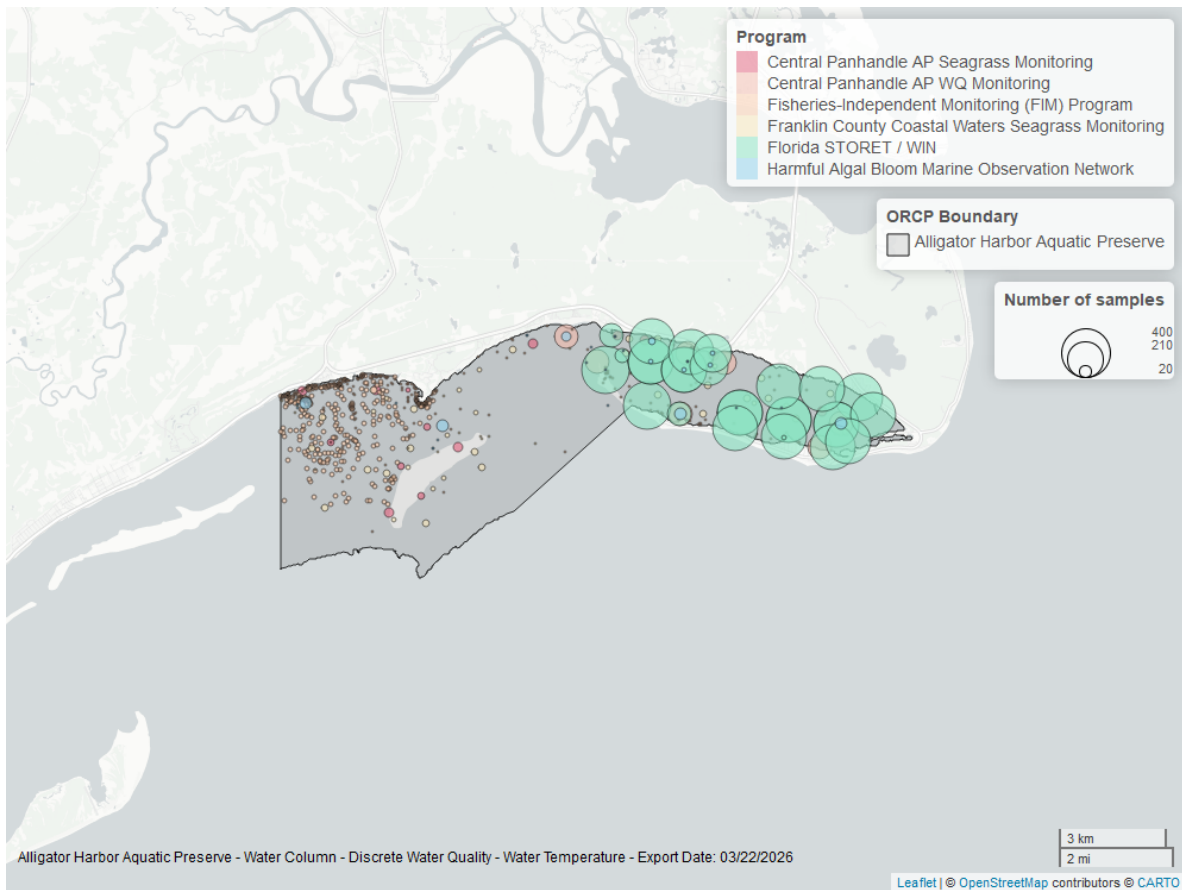


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

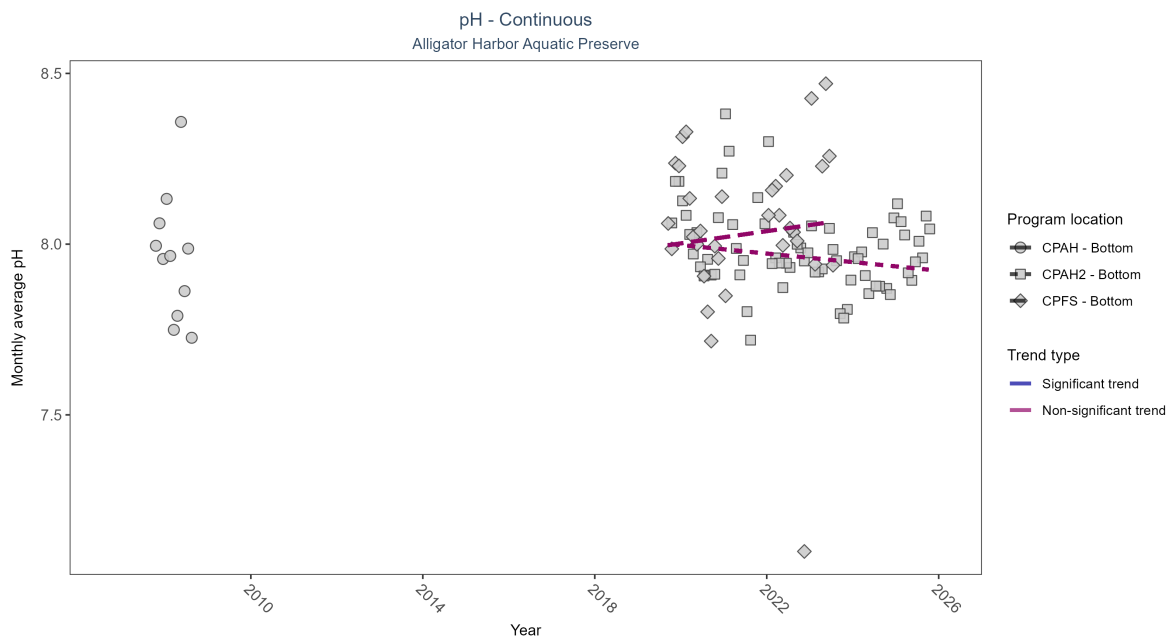


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data to calculate trend	13034	2	2007 - 2008	7.9	-	-	-	-
CPAH2	No significant trend	172942	7	2019 - 2025	8.0	-0.18	8.01	-0.01	0.094
CPFS	No significant trend	61868	5	2019 - 2023	8.1	0.19	7.98	0.02	0.5182

No detectable change in monthly average pH was observed at two locations. There was insufficient data to fit a model for one location.

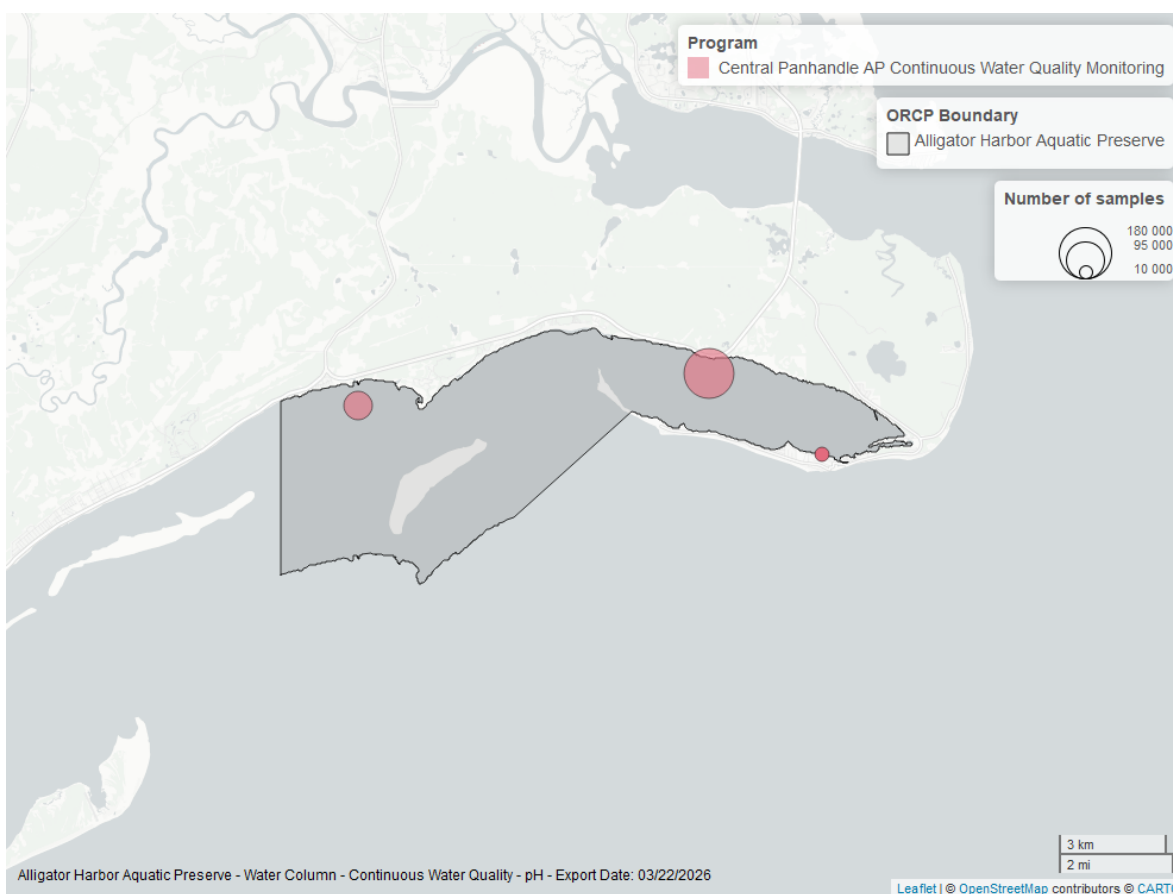


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

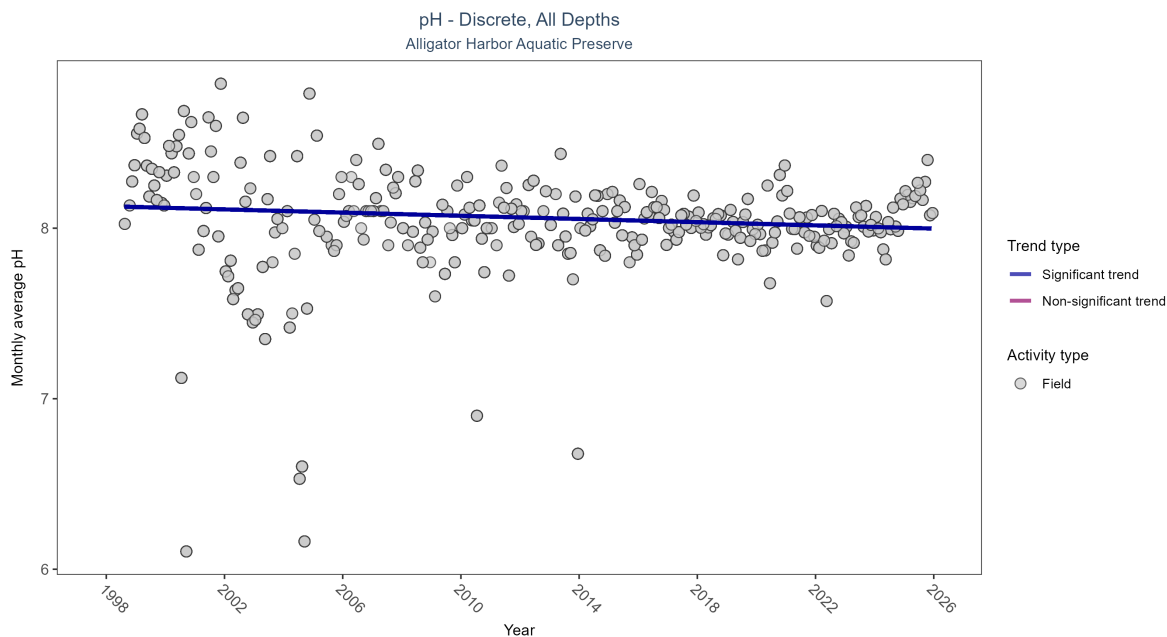


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	5534	28	1998 - 2025	8.1	-0.1142	8.129	-0.00469	0.0052

Monthly average pH decreased by less than 0.01 pH units per year.

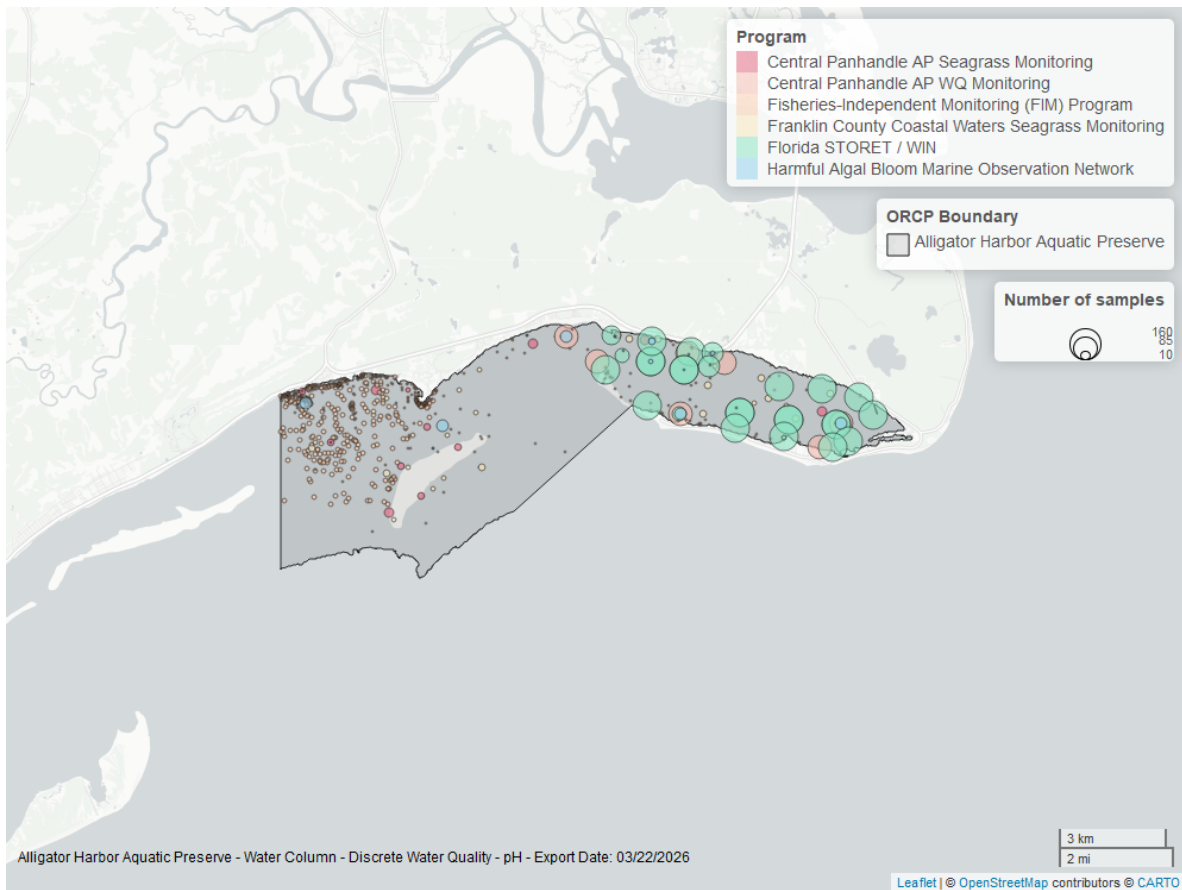


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

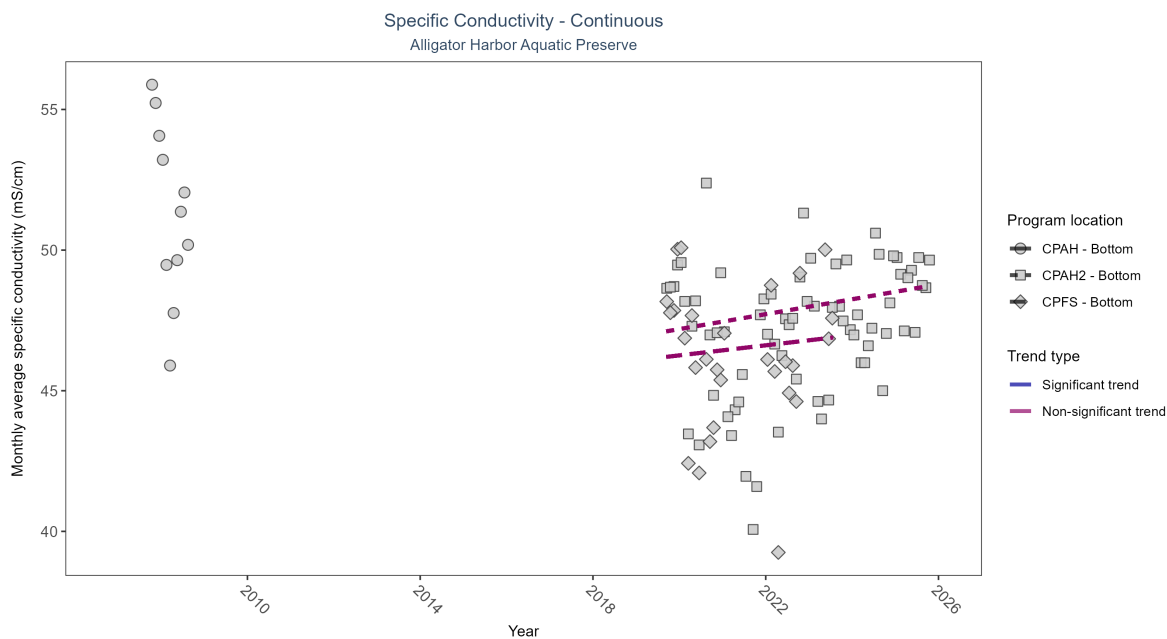


Table 13: Seasonal Kendall-Tau Results - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data to calculate trend	13034	2	2007 - 2008	51.89	-	-	-	-
CPAH2	No significant trend	171623	7	2019 - 2025	47.71	0.21	46.93	0.27	0.064
CPFS	No significant trend	60346	5	2019 - 2023	46.91	0	46.08	0.18	1

No detectable change in monthly average specific conductivity was observed at two locations. There was insufficient data to fit a model for one location.

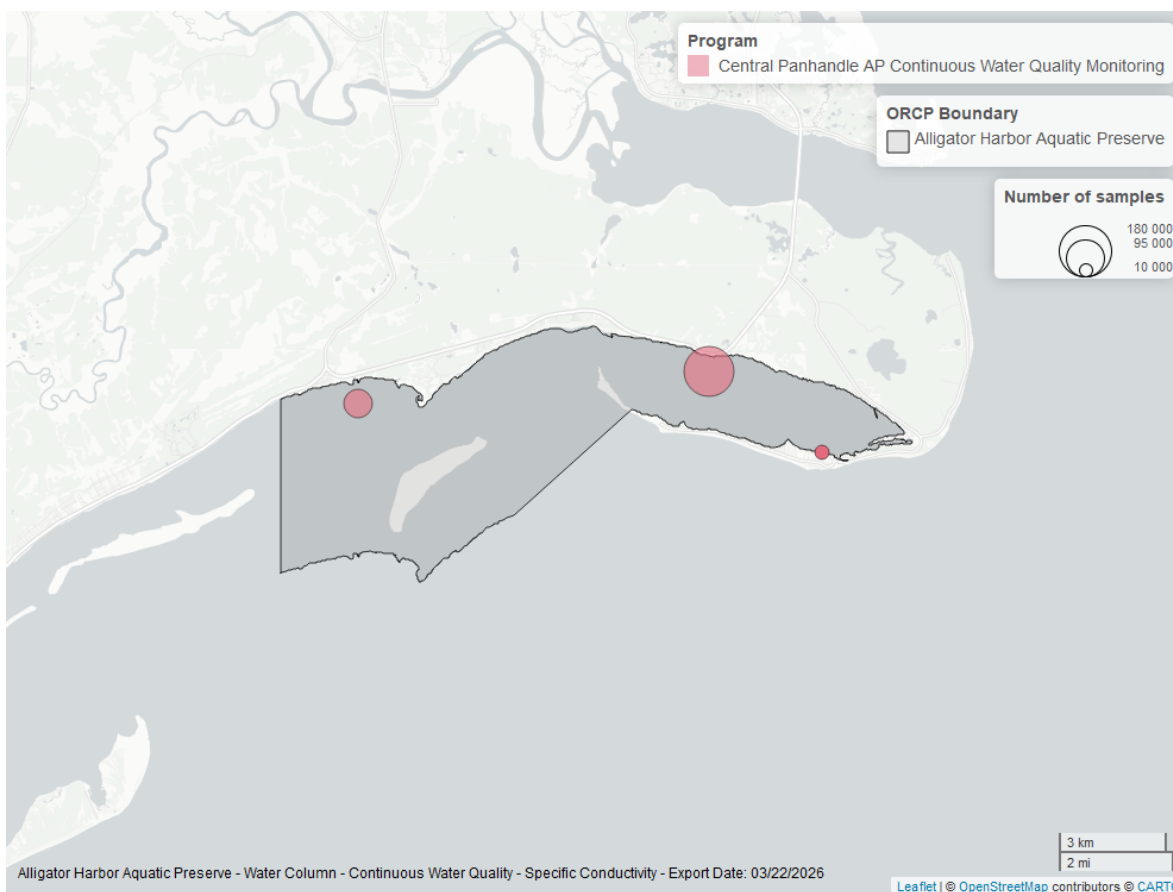


Figure 25: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

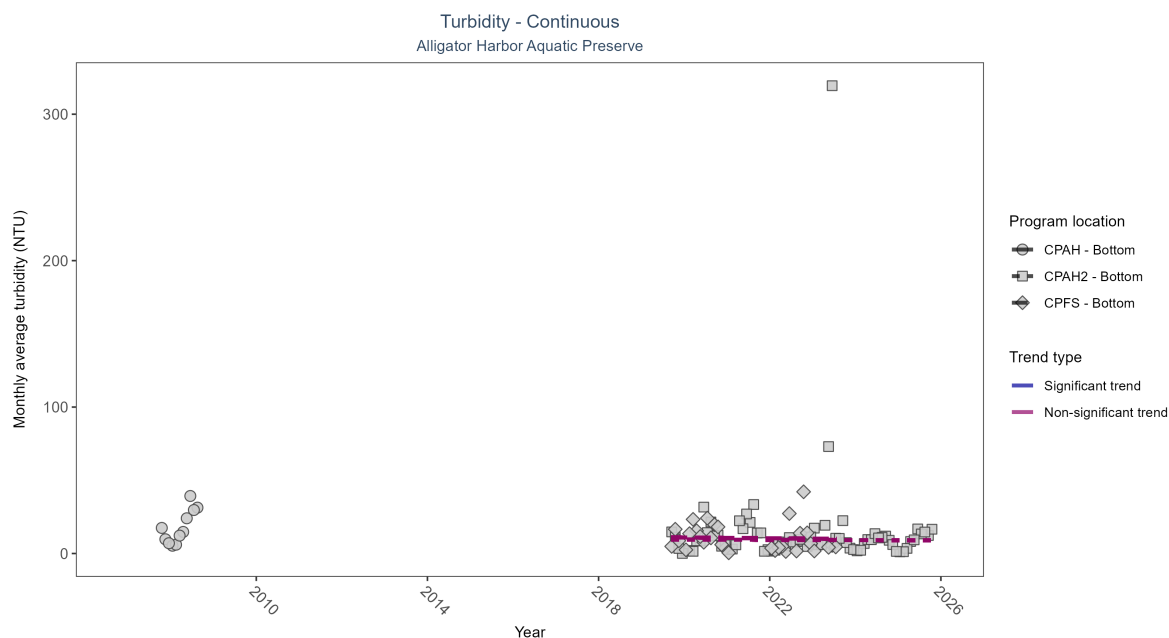


Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CPAH	Insufficient data to calculate trend	12558	2	2007 - 2008	10	-	-	-	-
CPAH2	No significant trend	176505	7	2019 - 2025	6	-0.03	9.58	-0.09	0.7955
CPFS	No significant trend	55980	5	2019 - 2023	3	-0.16	11.21	-0.26	1

No detectable change in monthly average turbidity was observed at two locations. There was insufficient data to fit a model for one location.

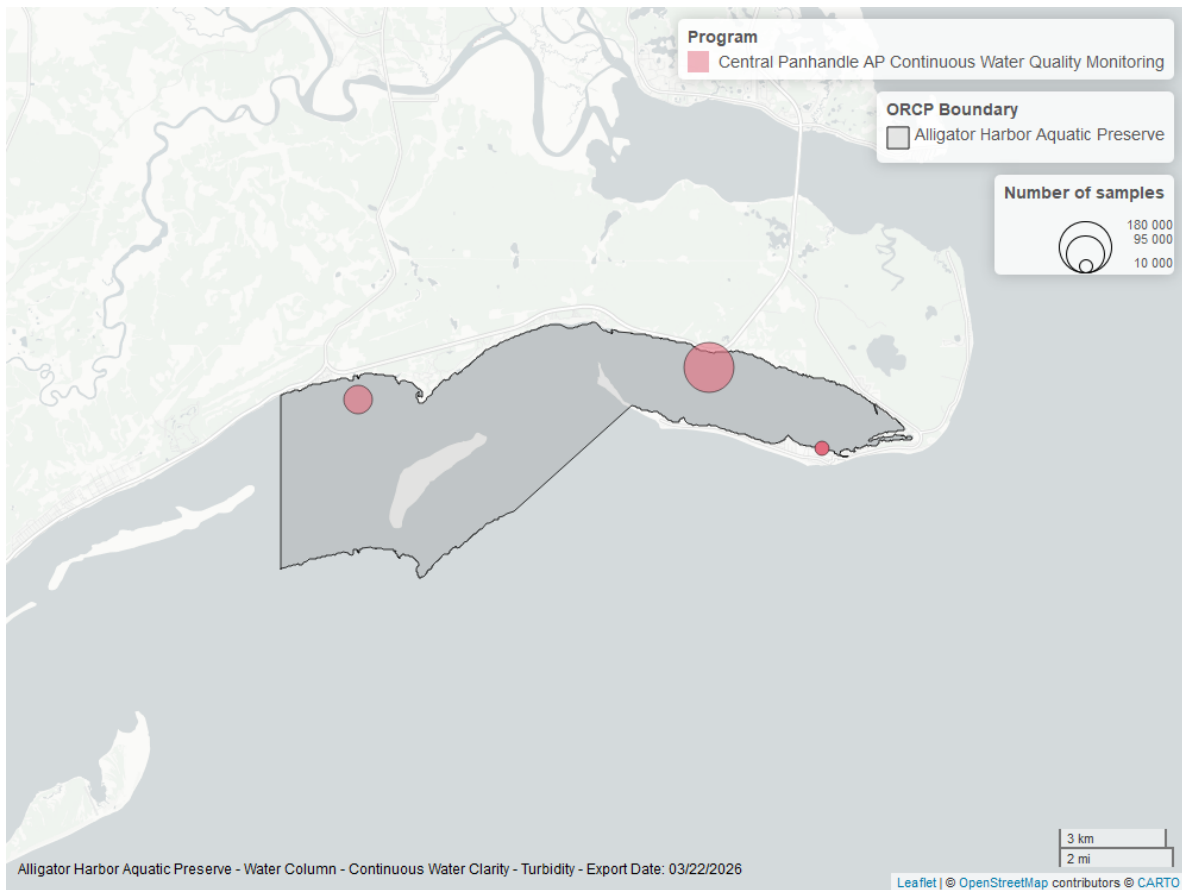


Figure 27: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

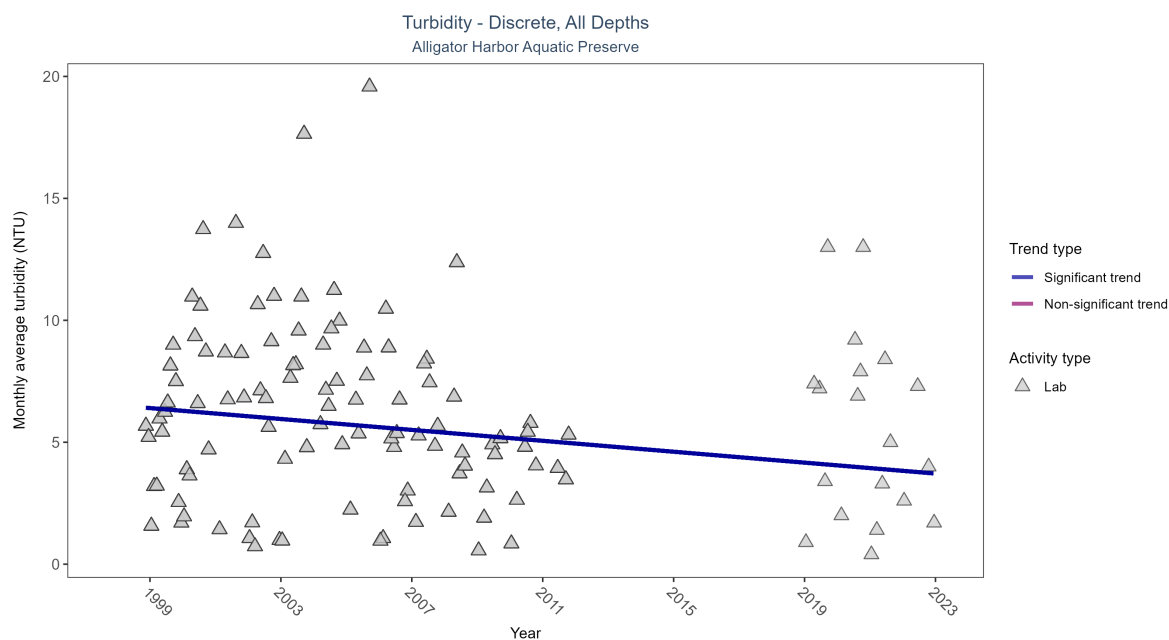


Figure 28: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	3543	18	1998 - 2022	5.3	-0.24321	6.51312	-0.11185	0.0012

Monthly average turbidity decreased by 0.11 NTU per year, indicating an increase in water clarity.

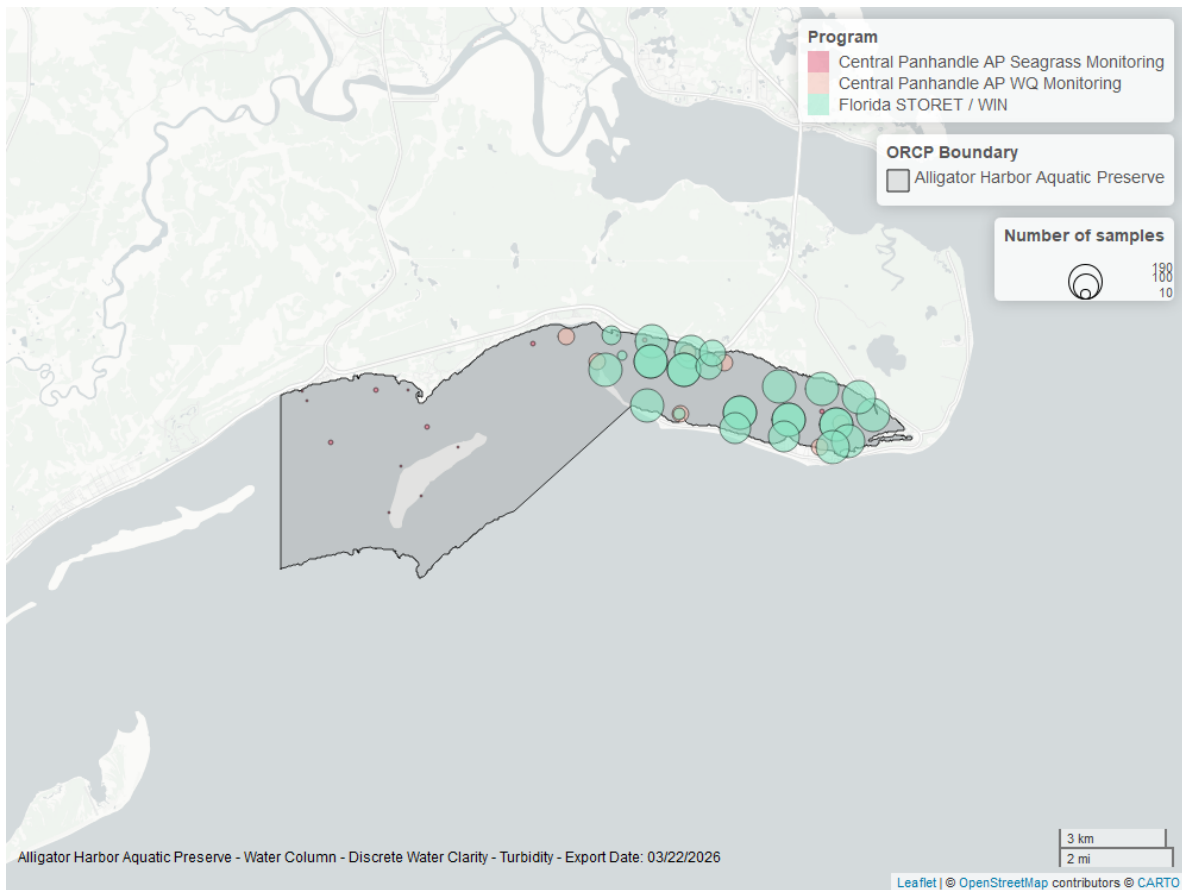


Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

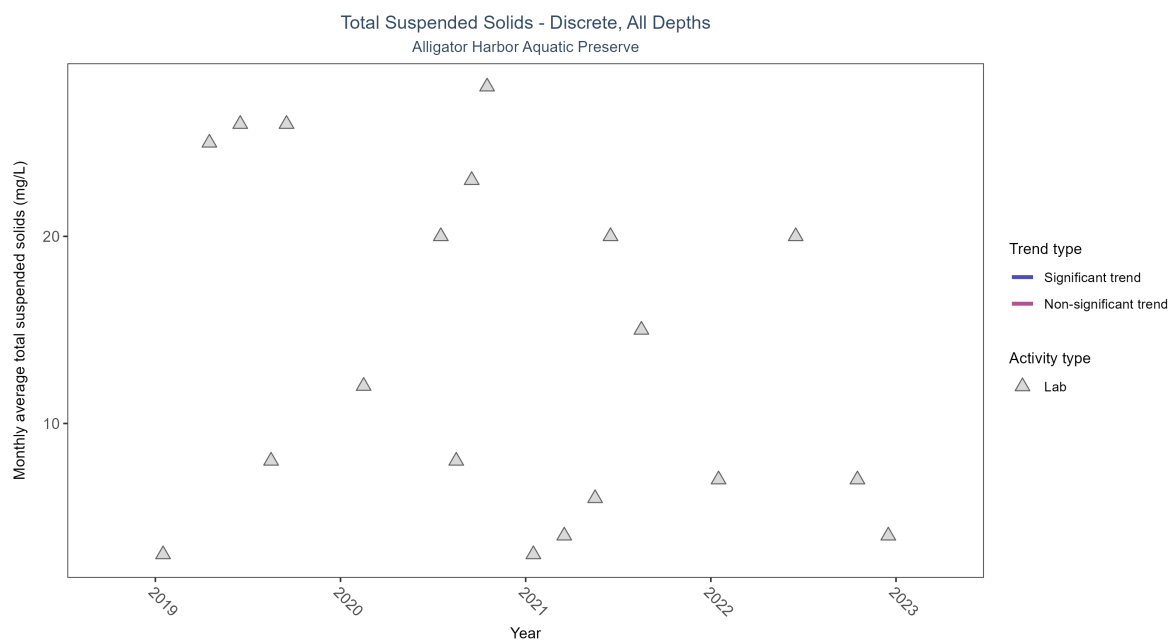


Figure 30: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data to calculate trend	19	4	2019 - 2022	12	-	-	-	-

There was insufficient data to fit a model for total suspended solids.

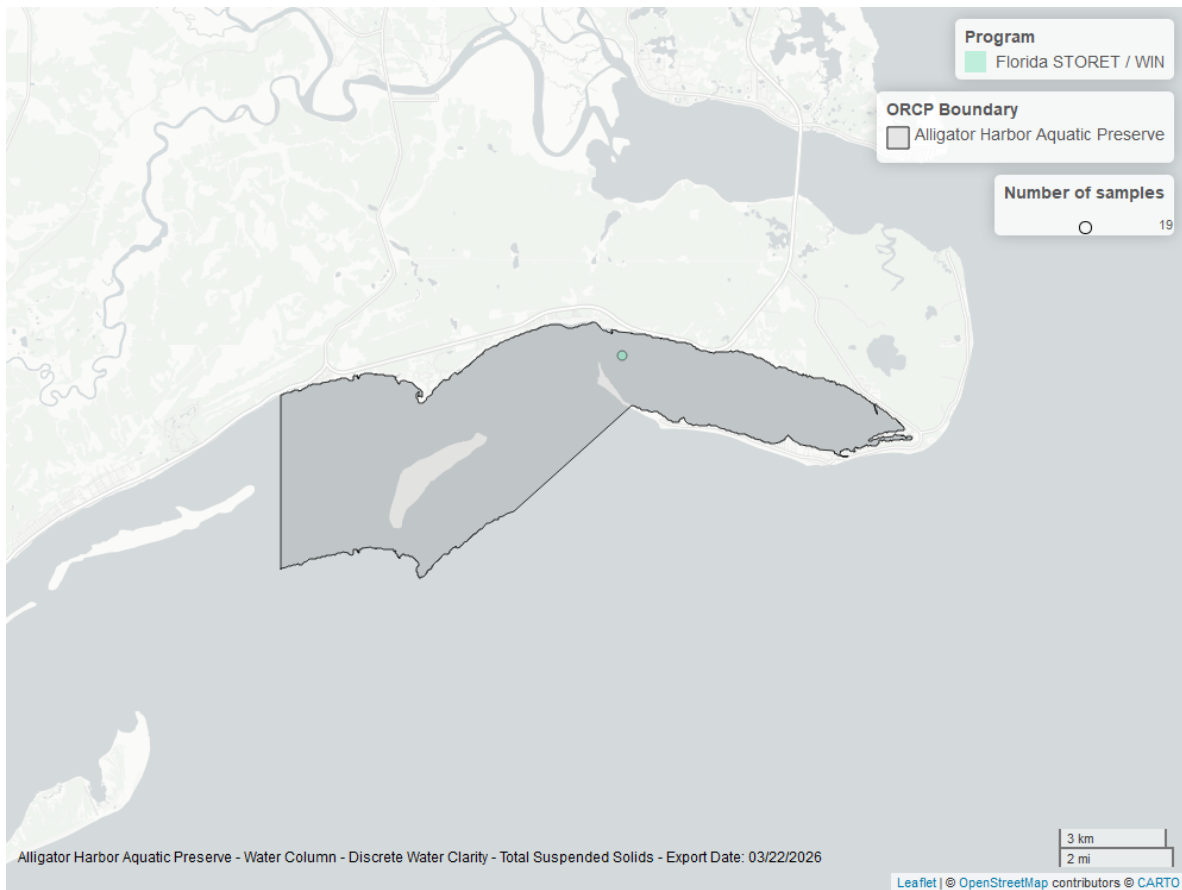


Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

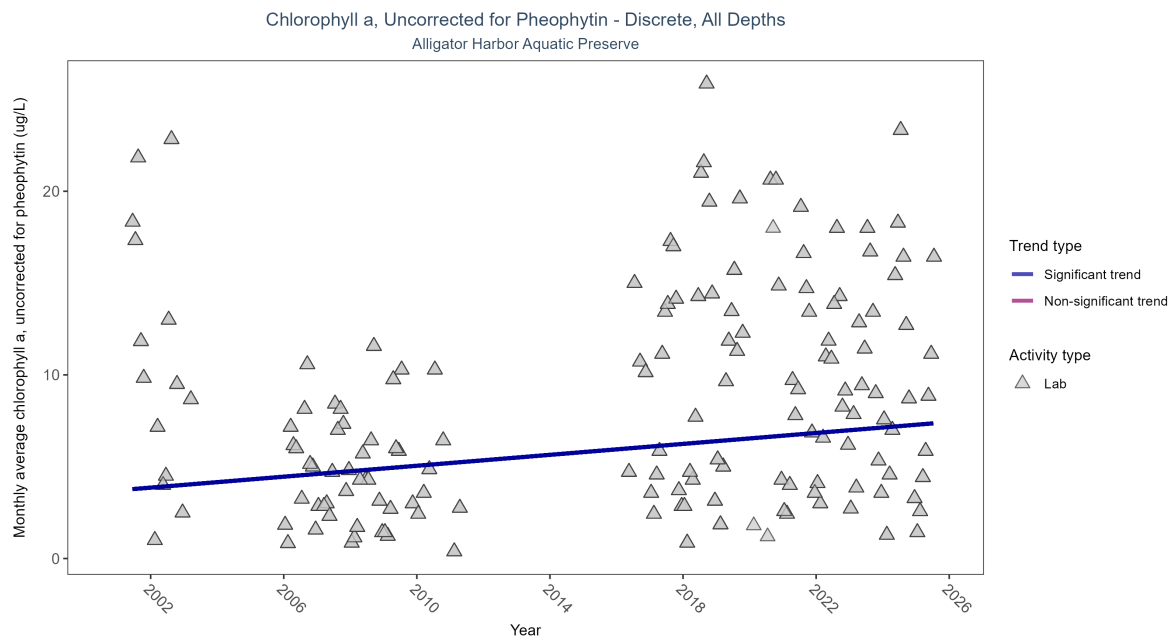


Figure 32: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	1178	19	2001 - 2025	6	0.27106	3.71199	0.14881	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.15 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

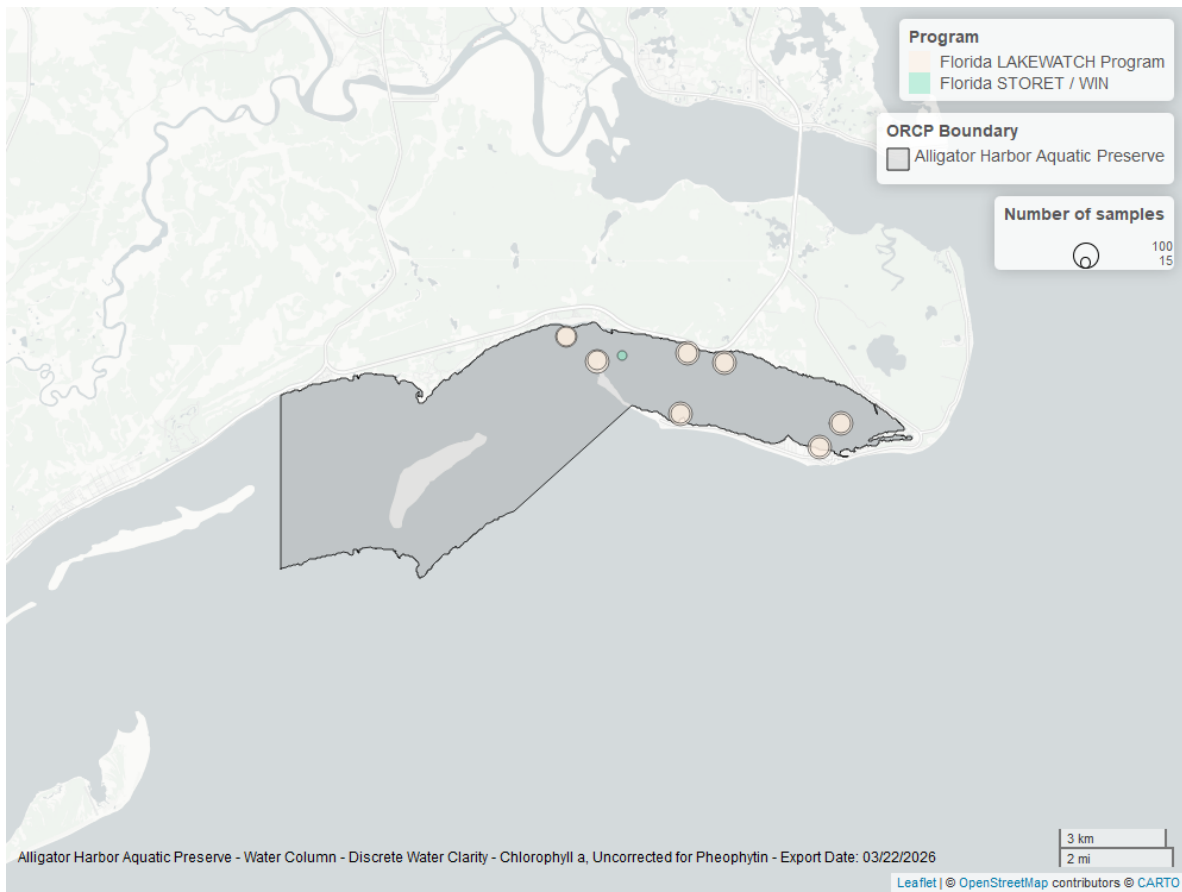


Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

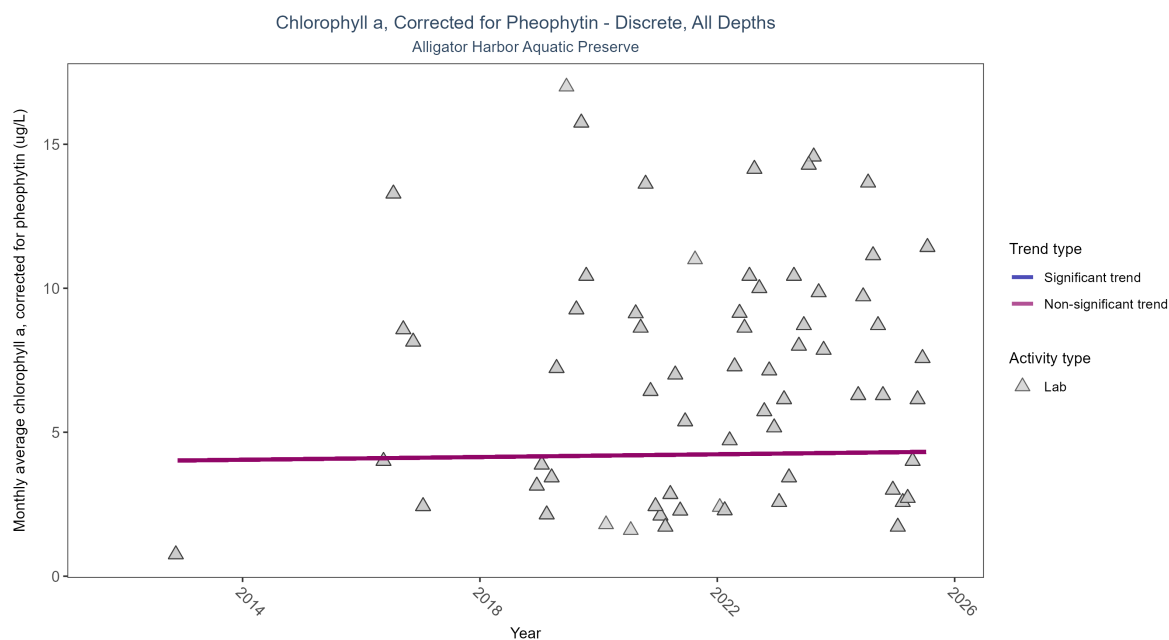


Figure 34: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	436	11	2012 - 2025	5.9	0.05897	3.99584	0.02381	0.5801

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2012 and 2025.

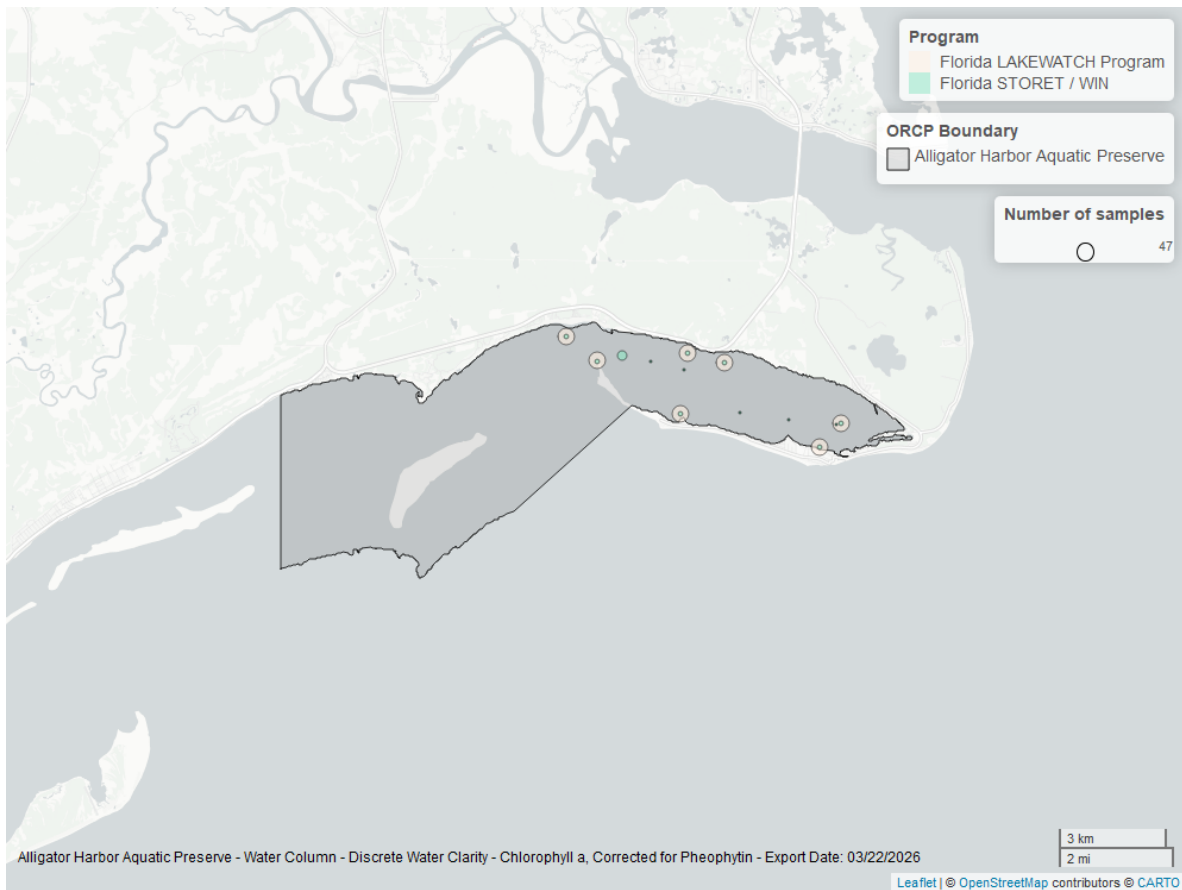


Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

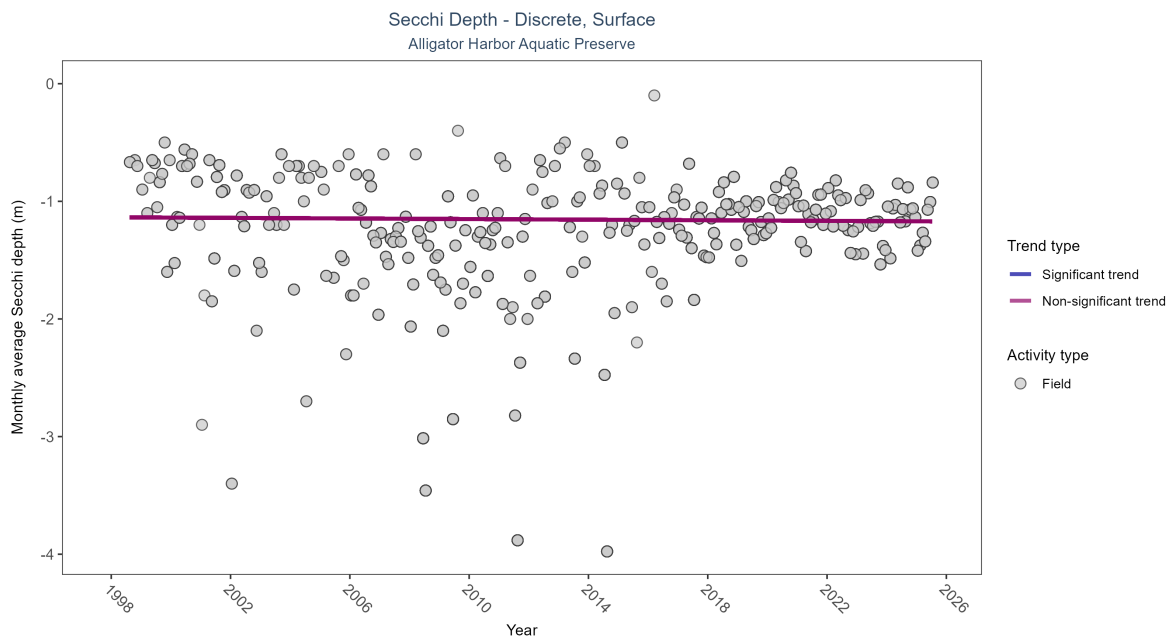


Figure 36: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	2596	28	1998 - 2025	-1.06681	-0.01209	-1.13577	-0.00126	0.7353

Secchi depth showed no detectable trend between 1998 and 2025.

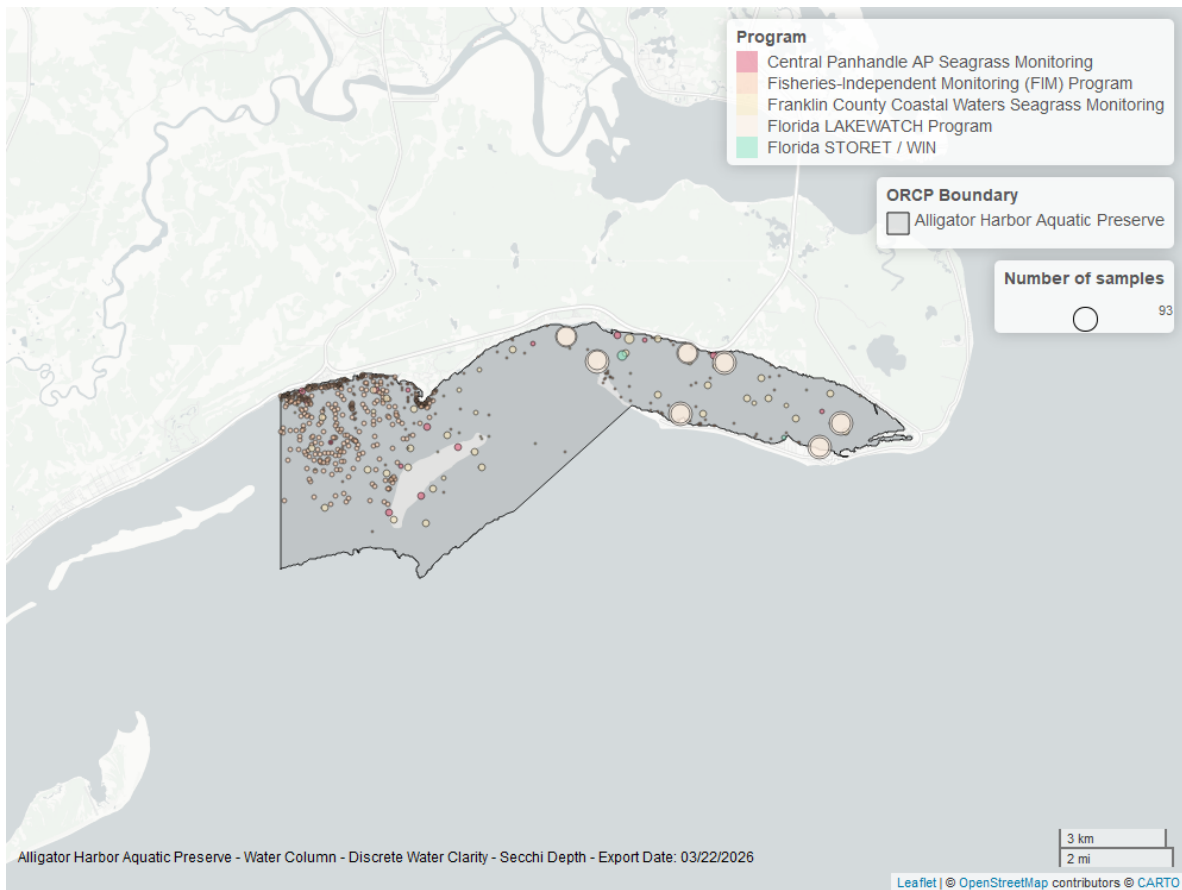


Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

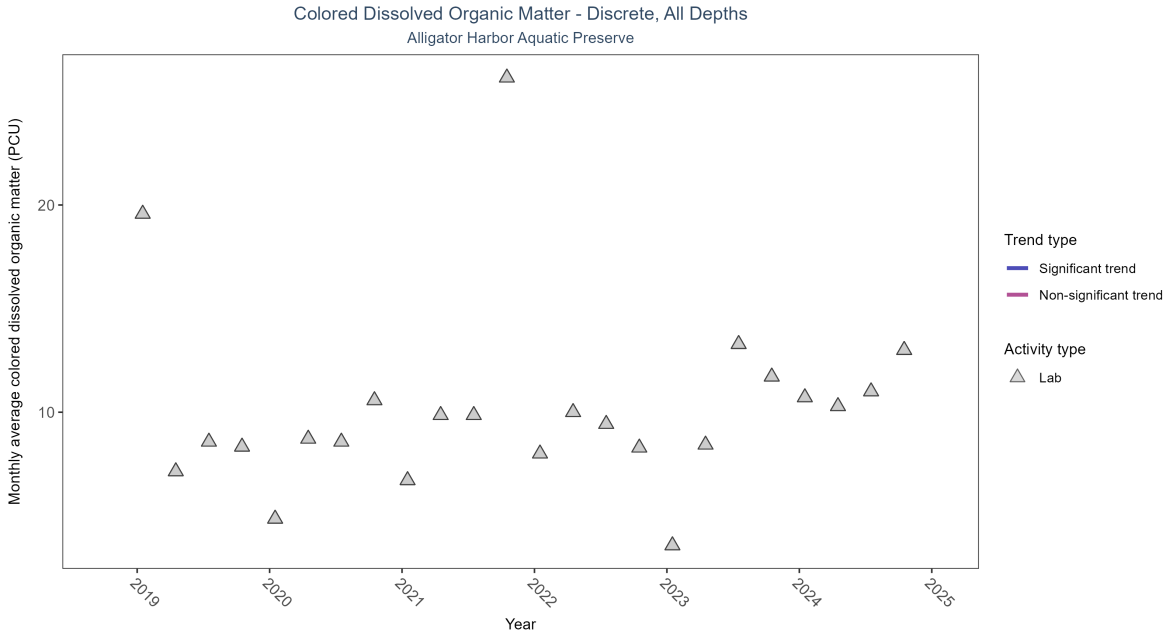


Figure 38: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data to calculate trend	167	6	2019 - 2024	9	-	-	-	-

There was insufficient data to fit a model for colored dissolved organic matter.

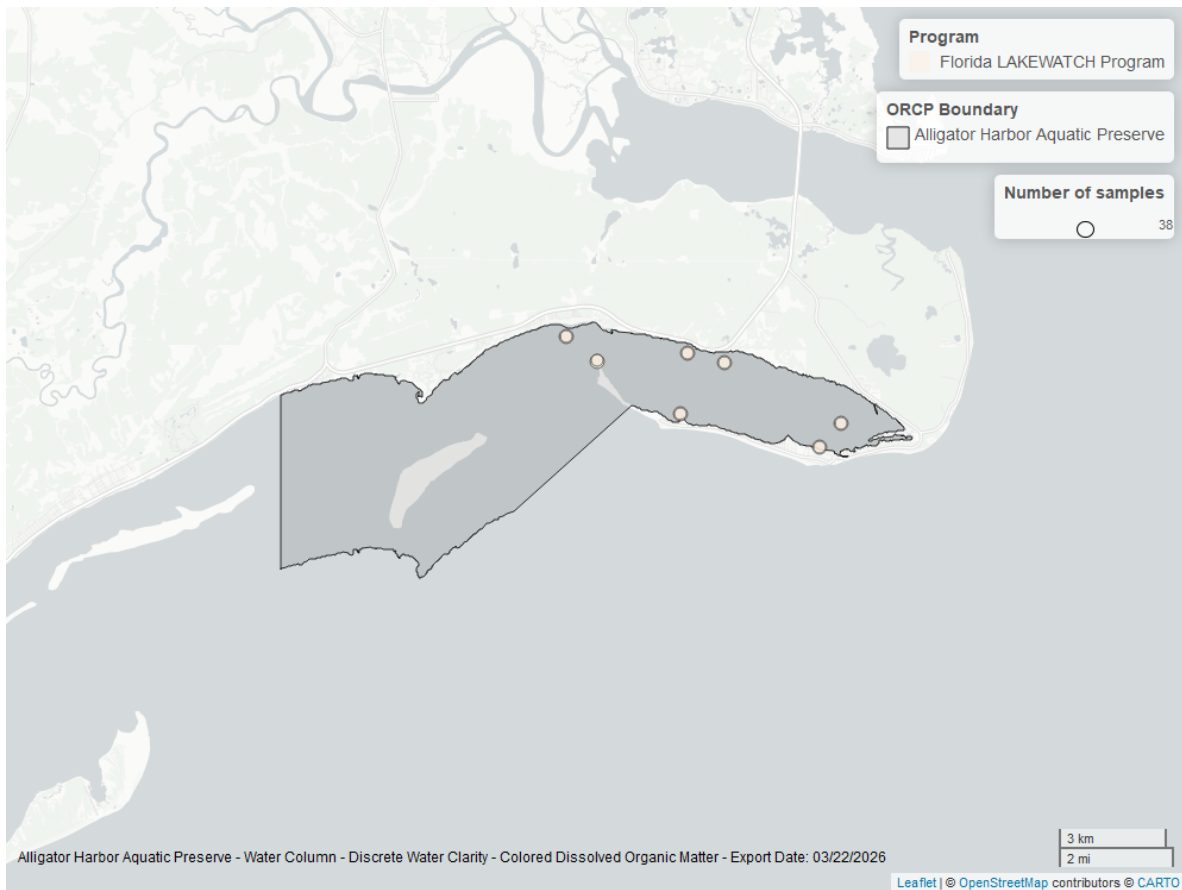


Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *Alligator Harbor Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.